

Inhibition of mitochondrial protein import and proteostasis by a pro-apoptotic lipid

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Abstract

Mitochondria mediated cell death is critically regulated by bioactive lipids derived from sphingolipid metabolism. The lipid aldehyde trans-2-hexadecenal (t-2-hex) induces mitochondrial dysfunction in a conserved manner from yeast to humans. Here we apply unbiased transcriptomic, functional genomics and chemoproteomic approaches in the yeast model to uncover the principal mechanisms and biological targets underlying this lipid-induced mitochondrial inhibition. We find that loss of Hfd1 fatty aldehyde dehydrogenase function efficiently sensitizes cells for t-2-hex inhibition and apoptotic cell death. Excess of t-2-hex causes a profound transcriptomic response with characteristic hallmarks of impaired mitochondrial protein import like activation of mitochondrial and cytosolic chaperones or proteasomal function and severe repression of translation. We confirm that t-2-hex stress induces rapid accumulation of mitochondrial pre-proteins and protein aggregates and subsequent activation of Hsf1- and Rpn4-dependent gene expression. By saturated transposon mutagenesis we find that t-2-hex tolerance requires an efficient heat shock response and specific mitochondrial and ER functions and that mutations in ribosome, protein and amino acid biogenesis are beneficial upon t-2-hex stress. We further show that genetic and pharmacological inhibition of protein translation causes t-2-hex resistance indicating that loss of proteostasis is the predominant consequence of the pro-apoptotic lipid. Several TOM subunits, including the central Tom40 channel, are lipidated by t-2-hex *in vitro* and mutation of accessory subunits Tom20 or Tom70 confers t-2-hex tolerance. Moreover, the Hfd1 gene dose determines the strength of t-2-hex mediated inhibition of mitochondrial protein import and Hfd1 co-purifies with Tom70. Our results indicate that transport of mitochondrial precursor proteins through the outer mitochondrial membrane is sensitively inhibited by the pro-apoptotic lipid and thus represents a hotspot for pro- and anti-apoptotic signaling.

eLife assessment

This study is a **valuable** observation that deals with the toxic effects of an intermediary in lipid degradation [trans-2-hexadecenal (t-2-hex)] in yeast through modification of mitochondrial protein import via the TOM complex. However, we find that the claim that the TOM complex is a main target of t-2-hex are supported by **incomplete** evidence, thus allowing multiple various interpretation. Despite the shortcomings, this study is inspiring for researchers from the organellar, protein trafficking and lipid field and serves as a starting point to further precise and mechanistic analyses of the phenomenon.

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Introduction

Mitochondria are universal and essential organelles, which beyond fundamental functions in energy metabolism, the biogenesis of Fe/S clusters and Ca^{2+} homeostasis, are the central regulators of the most common form of programmed cell death found in eukaryotic cells (Bock and Tait, 2020; Chipuk et al., 2021). Consequently, mitochondrial function and dysfunction critically establish cell survival or deterioration during aging, neurodegeneration and cancer (Area-Gomez et al., 2019; Porporato et al., 2018; Son and Lee, 2021). In this process of intrinsic apoptosis, cellular stress triggers pro-death signals, which converge at the organelle and induce the permeabilization of the outer mitochondrial membrane (MOMP). As a consequence, the release of several soluble mitochondrial proteins such as cytochrome *c* from the intermembrane space into the cytosol executes programmed cell death via caspase-dependent and -independent routes. Because MOMP is the earliest irreversible death trigger, its misregulation is causally linked to many diseases and a promising target for therapeutic interventions (Jeng et al., 2018). Accordingly, the mitochondria-mediated death decision needs to be strictly controlled. In higher eukaryotes, it is known that the entry into the death program depends on the functional interplay of different members of the BCL-2 protein family (Jeng et al., 2018). Specifically, BCL-2 protein BAX can transform under stress conditions from a latent cytosolic form to the pro-apoptotic oligomeric form, which initiates the perforation of the outer mitochondrial membrane and commit the cell to death (Walensky and Gavathiotis, 2011). One important induction mechanism of mitochondrial cell death is triggered by lipids derived from sphingolipid metabolism at the endoplasmic reticulum. Mitochondria devoid of ER membranes are resistant to pro-apoptotic inhibition, and specific lipid species alone, such as sphingosine-1-phosphate or trans-2-hexadecenal (t-2-hex), efficiently activate Bax-driven MOMP (Chipuk et al., 2012). The pro-apoptotic function of externally added t-2-hex has been generally confirmed in cultures from yeast to mammalian cells (Amaegberi et al., 2019; Kumar et al., 2011; Manzanera-Estreded et al., 2017). One explicit pro-apoptotic mechanism of t-2-hex has been recently discovered consisting in the direct lipidation and subsequent activation of Bax at C126 (Cohen et al., 2020). However, t-2-hex covalently and specifically modifies hundreds of different human proteins *in vitro* (Jarugumilli et al., 2018), which motivated our present study to identify biologically important t-2-hex targets in an exhaustive manner.

99% of the mitochondrial proteins are synthesized at cytosolic ribosomes and thus have to be imported into the organelle mostly post-translationally (Pfanner et al., 2019). This large variety of precursor proteins needs to pass through the mitochondrial membrane(s), sent to the correct mitochondrial subcompartment, be correctly folded and eventually assembled into protein complexes. Mitochondrial protein import is a proteostatic challenge for the cell and recent

research has revealed that multiple quality control mechanisms exist to avoid the accumulation of non-imported or aggregated precursor proteins (Song et al., 2021 [↗](#)). The multisubunit TOM complex (translocase of the outer membrane) is the entry channel for the vast majority of mitochondrial precursors. The β -barrel Tom40 protein forms the actual protein-transporting pore at the outer mitochondrial membrane (Hill et al., 1998 [↗](#)). The majority of the mitochondrial pre-proteins made in the cytosol has to be guided and maintained in an unfolded, import-competent state, which critically requires the function of cytosolic chaperones, such as Hsp70 and Hsp90 (Young et al., 2003 [↗](#)). This is important for cell homeostasis as prematurely folded pre-proteins cannot pass through and eventually clog the TOM complex, which leads to mitochondrial dysfunction and triggers a strong stress response (Boos et al., 2019 [↗](#)). Cellular rescue pathways, which prevent clogging of the TOM complex, have been recently discovered in yeast. The mitochondrial protein translocation-associated degradation pathway (mito TAD) continuously monitors protein import at the TOM complex upon normal growth conditions and removes stalled pre-proteins with the help of the ubiquitin binding protein Ubx2 and the AAA protein unfoldase Cdc48 by facilitating their proteasomal degradation (Mårtensson et al., 2019 [↗](#)). Whenever mitochondrial protein import is diminished by either mutation of translocase subunits, overexpression of clogging-prone pre-proteins or a loss of mitochondrial membrane potential, the mitochondrial compromised protein import response (mitoCPR) is activated. Upon prolonged mitochondrial import stress, the stress-induced Cis1 protein recruits the Msp1 AAA ATPase at the TOM complex to remove arrested precursor proteins by proteasomal degradation (Weidberg and Amon, 2018 [↗](#)). In situations when these internal rescue systems are overwhelmed, the accumulation of non-imported mitochondrial precursor proteins in the cytosol poses a serious threat to cellular proteostasis and causes a strong proteotoxic stress (Wang and Chen, 2015 [↗](#); Wrobel et al., 2015 [↗](#)). Under these conditions, the cell responds with a massive transcriptional remodeling, which involves heat shock factor 1 (Hsf1) and Rpn4 in order to induce the expression of cytosolic chaperones and the ubiquitin proteasome system, respectively (Boos et al., 2019 [↗](#)). The efficiency of mitochondrial protein import is of central importance for general cytosolic protein folding and homeostasis (Boos et al., 2020 [↗](#)). It has been very recently demonstrated that the toxic aggregation of polyQ, amyloid β and α -synuclein proteins relevant for neurodegenerative and aging phenotypes is dependent on the efficiency of mitochondrial protein import (Nowicka et al., 2021 [↗](#); Schlagowski et al., 2021 [↗](#)). Here we demonstrate that mitochondrial protein import is inhibited by the naturally occurring pro-apoptotic sphingolipid degradation intermediate t-2-hex, the accumulation of which leads to a massive proteostatic response in the cytosol to prevent general protein aggregation. Our results connect pro-apoptotic lipid function with protein misfolding and aggregation via a block in mitochondrial protein import.

Results

The stress-regulated fatty aldehyde dehydrogenase Hfd1 is the major detoxifier of pro-apoptotic t-2-hex

Intracellular production of the pro-apoptotic lipid t-2-hex occurs in eukaryotic cells via the evolutionarily conserved sphingolipid degradation pathway (**Figure 1A** [↗](#)). In human cells, t-2-hex causes mitochondrial dysfunction by directly stimulating Bax oligomerization at the outer mitochondrial membrane. In yeast, however, t-2-hex efficiently interferes with mitochondrial function and cell growth in a Bax independent manner (Manzanares-Estreded et al., 2017 [↗](#)). Thus, we wanted to identify novel lipid-mediated apoptotic mechanisms in the yeast model. In *S. cerevisiae*, the degradation of sphingosine to palmitic acid occurs through enzymatic steps, which seem to be transcriptionally activated upon cellular stress (Manzanares-Estreded et al., 2017 [↗](#)). The fatty aldehyde dehydrogenase Hfd1 acts as a potential direct detoxifier by oxidizing t-2-hex. We analyzed the regulation of *HFD1* by cytotoxic stresses with time-elapsd luciferase assays *in vivo*. We applied different treatments including NaCl for the induction of osmotic stress, menadione for the induction of oxidative stress (Flattery-O'Brien et al., 1993 [↗](#)) and acetic acid for

the induction of pro-apoptotic stress (Chaves et al., 2021 [↗](#)). Indeed, *HFD1* expression was fast, robustly, and transiently induced by the three types of stress (**Figure 1B** [↗](#)). The sensitivity of the observed activation was comparable to that of a generic stress-inducible gene, *GRE2*, except for oxidative stress, where *HFD1* showed a less sensitive response (**Figure 1C** [↗](#)). This suggested that the yeast sphingolipid degradation pathway is generally induced by diverse stresses and that increased Hfd1 levels might be necessary to reduce toxic t-2-hex levels under all these conditions. We next asked whether increased t-2-hex levels itself triggered at all a transcriptional stress response in yeast cells. External addition of the bioactive lipid indeed caused a dose-dependent activation of the *HFD1* and *GRE2* genes (**Figure 1D** [↗](#)). Of note, in contrast to the transient stimulation of gene expression caused by “conventional” abiotic stresses (**Figure 1B** [↗](#)), t-2-hex overload activated gene expression in a much more sustained manner. This suggested that the lipid caused intracellular damage, which builds up slowly and affects cell function over a prolonged time. We tested whether the amount of Hfd1 enzyme was critical for the extent of t-2-hex inflicted damage. We compared how the sensitivity to the lipid changed upon absent or constitutively high Hfd1 activity. *hfd1Δ* cells exhibited a very pronounced growth delay upon t-2-hex treatment, while constitutively Hfd1 overexpressing cells gained robust growth upon the same conditions (**Figure 1E** [↗](#)). These results confirmed that *HFD1* gene dose and enzyme activity were critical determinants for the tolerance to the pro-apoptotic lipid. Finally, we found that the loss of Hfd1 function greatly enhanced the adaptive activation of gene expression (**Figure 1F** [↗](#)). Furthermore, *hfd1* mutants proliferated less efficiently upon pro-apoptotic stress conditions, while cells with constitutive *HFD1* were slightly resistant (**Figure 1G** [↗](#)). These data suggest that the growth arrest by pro-apoptotic t-2-hex is modulated by the efficiency of internal t-2-hex detoxification, which depends on the Hfd1 enzyme levels. *hfd1Δ* cells are highly sensitized to t-2-hex toxicity exhibiting a several fold enhanced response, a fact that will be exploited next to discover the complete transcriptomic response to the pro-apoptotic lipid.

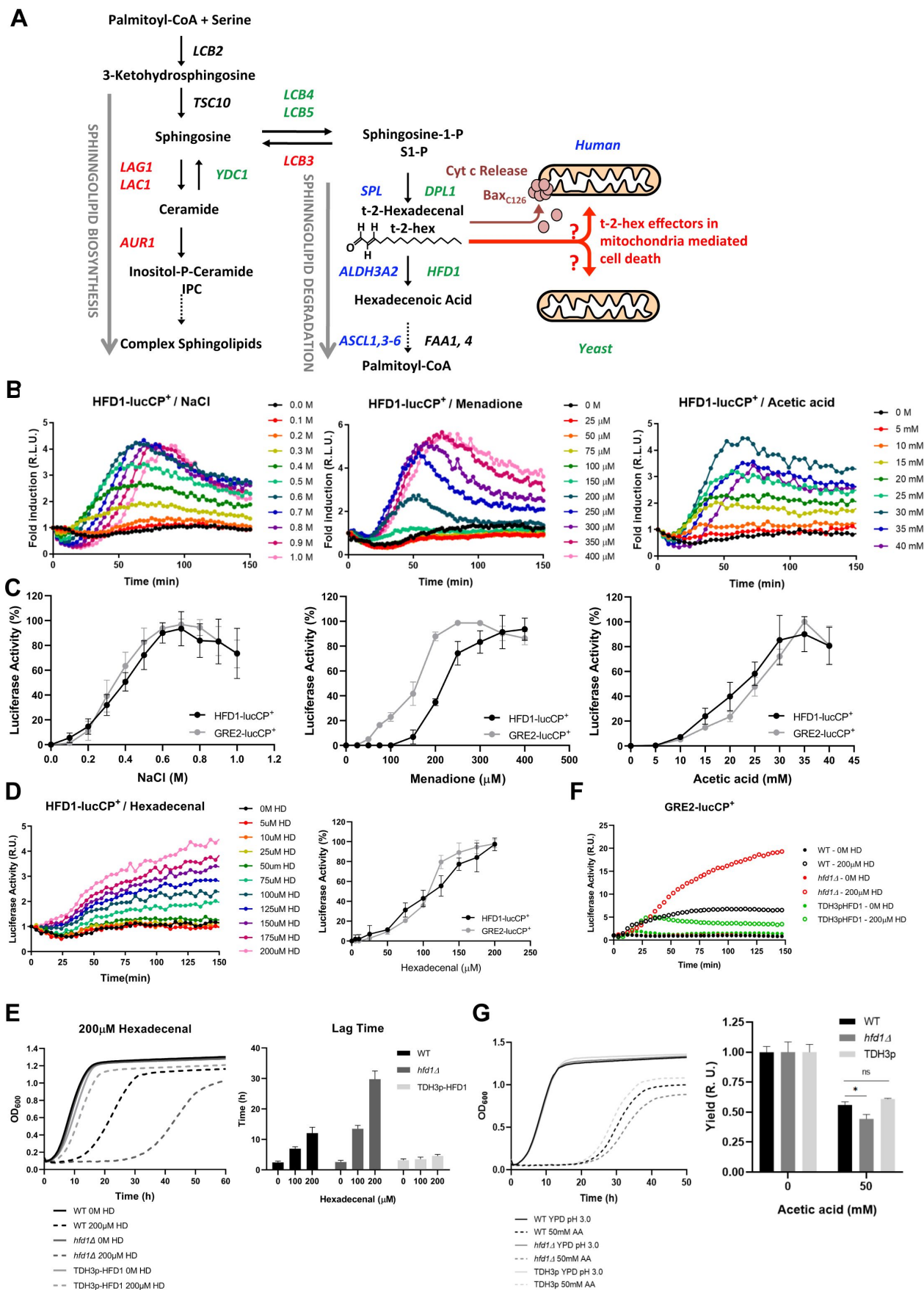


Figure 1.

The stress-activated fatty aldehyde dehydrogenase Hfd1 determines t-2-hex mediated toxicity and adaptive response. (A) Overview of the generation of t-2-hex within the sphingolipid degradation pathway. Stress-activated (-repressed) enzymatic functions are depicted in green (red) for yeast. The corresponding human enzymes involved in t-2-hex chemistry are shown in blue. t-2-hex stimulates the pro-apoptotic activity of Bax at human mitochondria by C₁₂₆ lipidation. (B) *HFD1* expression is activated by different cytotoxic stresses. Hyperosmotic (NaCl), oxidative (Menadione) and pro-apoptotic (Acetic acid) stresses were gradually applied to *HFD1p*-luciferase reporter containing yeast cells. The relative luciferase activity was measured *in vivo* (n = 3). (C) Comparison of different stress sensitivities of the *HFD1* and *GRE2* promoters. (D) External addition of t-2-hex (HD) activates *HFD1* expression in a concentration dependent manner. Upper panel: *HFD1*-luciferase activation upon gradual addition of t-2-hex (n = 3). Lower panel: Comparison of *HFD1* and *GRE2* activation sensitivities to t-2-hex. (E) Left panel: *HFD1* gene dose modulates susceptibility to t-2-hex. Growth curves of wild type, *hfd1Δ* and constitutively Hfd1 overexpressing strains (*TDH3p-HFD1*) in the absence and presence of t-2-hex (n = 3). Right panel: Quantitative comparison of growth performance (length of lag phase) of the same strains upon the indicated t-2-hex concentrations (n = 3). (F) Loss of Hfd1 function causes a hypersensitive t-2-hex response. *GRE2p*-luciferase reporter assay in strains with the indicated *HFD1* gene dose in response to t-2-hex (n = 3). (G) *HFD1* gene dose modulates the tolerance to pro-apoptotic concentrations of acetic acid (n = 3). *p<0.05 by Student's unpaired t-test.

The transcriptomic response to t-2-hex reveals global features of proteostatic imbalance

Our results indicated that cells lacking Hfd1 activity remain for many hours in a non-proliferative state when challenged with a t-2-hex overload. In this period of time, a strong regulation of gene expression occurs, presumably in order to prepare defense mechanisms against the toxic effects of the lipid. We were interested in capturing this adaptive response and determined the transcriptomic changes upon t-2-hex addition in *hfd1Δ* cells by RNA-seq (Figure 2A). We found that adaptation to the pro-apoptotic lipid involved the profound remodeling of the yeast transcriptome. The expression of approximately 30% of the transcriptome was more than 3 fold up- or down-regulated upon t-2-hex stress (Supplemental Tables 1 and 2). Very strong (> 10 fold) activation was observed for a remarkably high number of protein encoding genes (263), which were preferentially distributed in categories such as “stress-related enzyme function”, “mitochondrial function”, “transport function”, or “chaperones”. The most repressed gene functions fell into the categories “ribosome and translation”, “pheromone response and mating”, “nucleotide biosynthesis”, or “cell cycle”. Next, genes up-regulated (>2 log₂FC) or down-regulated (<-2 log₂FC) by t-2-hex were selected and subjected to GO category enrichment analysis (up-regulated, Supplemental Table 1; down-regulated, Supplemental Table 2). We found that “Mitochondrial organization” was the most numerous GO group activated by t-2-hex, while it was “Ribosomal subunit biogenesis” for t-2-hex repression. We therefore analyzed the most strongly t-2-hex induced genes encoding mitochondrial proteins (Figure 2B). Remarkably, *CIS1* showed an extraordinarily high induction (> 250 fold) and thus represented the most inducible gene with a known mitochondrial function. Cis1 is the central protein of the MitoCPR pathway, which is specifically induced upon and protects from mitochondrial protein import stress, together with the Msp1 AAA-ATPase (Weidberg and Amon, 2018). *MSP1* expression, although more moderately, is also significantly induced upon t-2-hex stress. These data suggested that mitochondrial protein import might be impaired by the action of the pro-apoptotic lipid. Accordingly, we found the expression of a large group of mitochondrial chaperones to be significantly activated upon t-2-hex exposure (Figure 2B). We compared our t-2-hex transcriptomic response with the previously described specific response to mitochondrial protein import block (Weidberg and Amon, 2018), taking into account only robustly (> 3 fold) induced genes. We found that 69% of all genes induced by mitochondrial import stress were also positively affected by t-2-hex stress (Supplemental Table 3). The number of coinciding genes increased to

85%, when we exclusively compared mitochondrial functions in both datasets, indicating that the response to the bioactive lipid included a specific adaptation to a failure in the mitochondrial protein import.

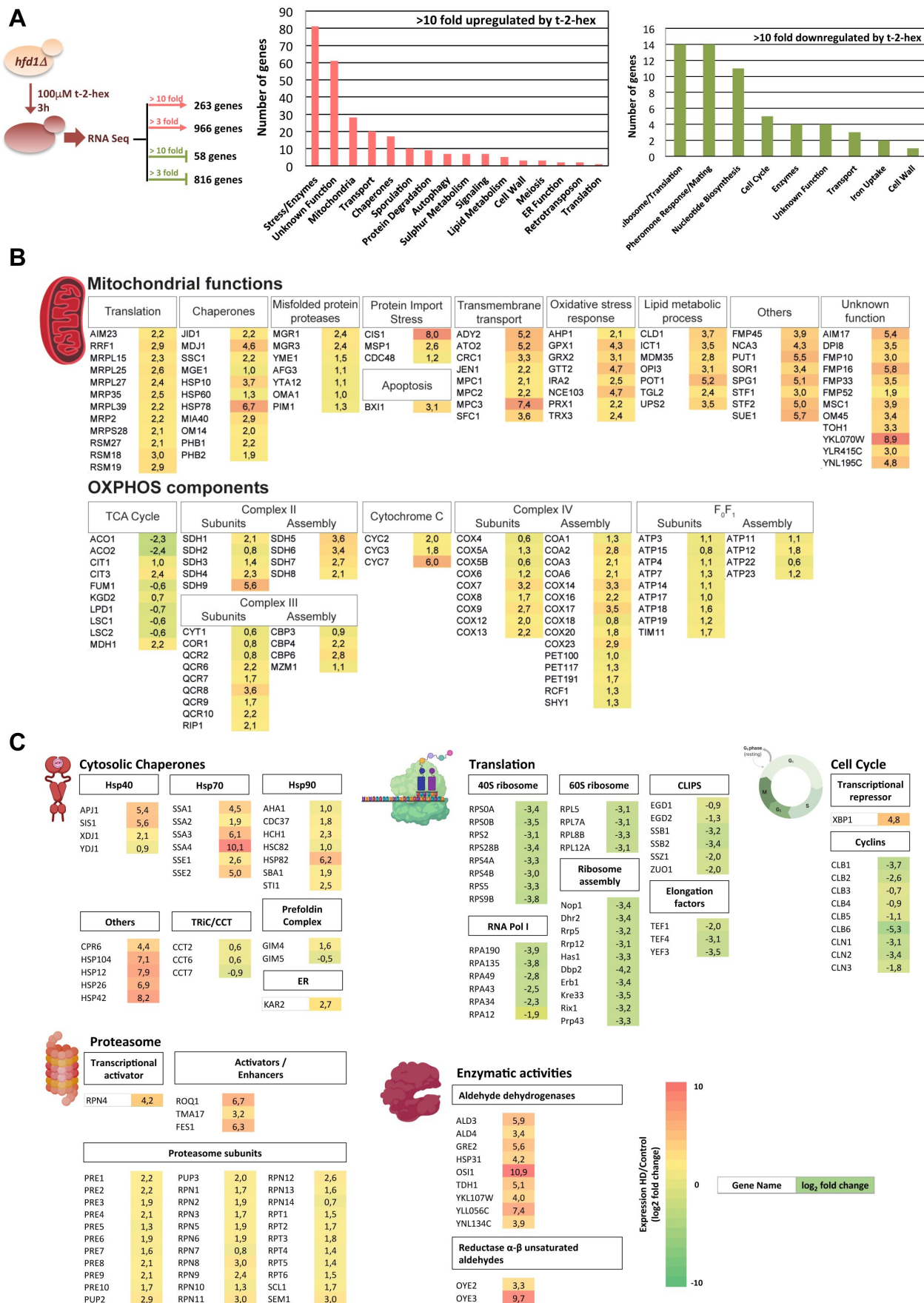


Figure 2.

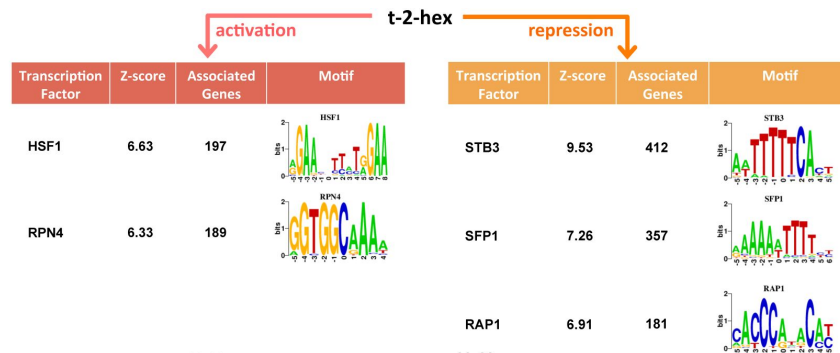
t-2-hex stress causes profound transcriptomic remodeling. (A) Transcriptomic analysis of the cellular response to t-2-hex overload. RNA Seq experiment setup and general functional groups of up- and down-regulated genes. (B) t-2-hex activated mitochondrial functions. (C) Genomic remodeling of gene expression reveals a general proteostatic response of the cell to t-2-hex. Cytosolic chaperones are strongly activated in general, but not the Cct chaperonin or the prefoldin complex. Structural components and regulators of the proteasome are coordinately up-regulated, while the cytosolic translation machinery and cell cycle regulators are strongly repressed. Significantly up-regulated enzymes with known aldehyde dehydrogenase or unsaturated aldehyde reductase activities are shown. Colors indicate $\log_2$ fold changes in gene expression of t-2-hex treated versus mock treated cells (means of $n = 3$ independent biological replicates) for genes representing selected functional groups or complexes.

The accumulation of non-imported mitochondrial proteins triggers a global transcriptional proteostatic response (Boos et al., 2019). We therefore investigated whether the t-2-hex response contained general features of this proteostatic adaptation. Strikingly, the expression of many cytosolic chaperone systems of the Hsp70, Hsp90, Hsp40 and other groups was highly activated by t-2-hex overload (Figure 2C), with some chaperone genes showing extraordinary induction levels of 100 to 1000-fold. Notably, these chaperones are key players in the maintenance of proteostasis in the cell (Sontag et al., 2017). The expression of other chaperone systems, such as the Cct chaperonin or the prefoldin complexes, dedicated to the correct assembly of cytoskeleton components, is not significantly affected by the lipid aldehyde (Figure 2C). Additionally, we observed a remarkably homogeneous upregulation of the expression of all proteasomal subunits and strong activation of positive proteasomal regulators, such as the Rpn4 transcriptional activator of proteasomal genes or the Roq1, Fes1 and Tma17 stimulators of proteasomal assembly and activity (Gowda et al., 2013; Hanssum et al., 2014; Szoradi et al., 2018). In parallel, our RNA seq data revealed that t-2-hex repressed very strongly the expression of genes involved in the *de novo* biosynthesis of ribosomes and proteins. This inhibition was observed in a highly consistent manner for specific RNA polymerase I and ribosomal subunits, as well as ribosome assembly factors, translational elongation factors and co-translationally acting chaperones (CLIPs). These data indicated that adaptation to the pro-apoptotic t-2-hex involved the activation of cytosolic chaperone systems and the proteasomal protein degradation machinery, as well as a general repression of translation, presumably as a consequence of imbalances in the cytosolic proteostasis caused by impaired mitochondrial protein import. This seems to be accompanied by a general arrest of cell cycle progression, because we observed strong repression of all cyclin genes and simultaneous up-regulation of the Xbp1 transcriptional master regulator of quiescence and cell cycle repression (Figure 2C).

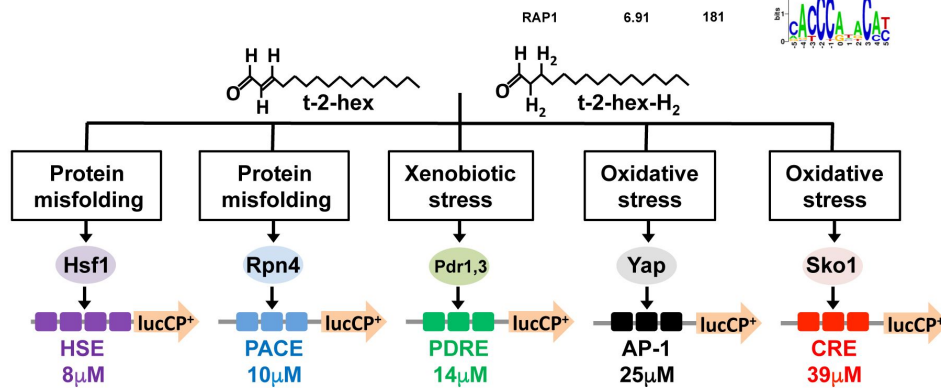
In order to further characterize the gene groups, which were most significantly affected by t-2-hex stress, we examined the significantly up- and down-regulated promoters by *in silico* analysis for enriched transcription factor binding motifs. Within the t-2-hex induced promoter regions, we found a consistent and strong enrichment of binding sites for Hsf1 and Rpn4 (Figure 3A). In yeast, Hsf1 is the essential master regulator of proteostasis, controlling both basal and heat stress induced expression of many chaperone encoding genes by binding to heat shock promoter elements (HSE) (Akerfelt et al., 2010). Rpn4 is a non-essential, general activator of proteasome genes, which binds to so called proteasome-associated control elements (PACE) (Xie and Varshavsky, 2001). In turn, for the t-2-hex-repressed gene promoters, we identified the binding motifs for Stb3, Sfp1 and Rap1 as the most significantly enriched regulatory elements (Figure 3A). Stb3 is a general activator of yeast ribosomal RNA genes and binds to the ribosomal RNA processing elements (RRPE) (Liko et al., 2007). Sfp1 and Rap1 stimulate the expression of ribosomal protein and biogenesis genes (Lieb et al., 2001; Marion et al., 2004). These data suggested that t-2-hex triggered the upregulation of chaperone and proteasome genes via the Hsf1 and Rpn4 transcriptional activators as the main adaptive response, while it massively repressed

ribosomal biogenesis. We wanted to experimentally prove the prevalence of the heat shock and proteasomal response in the case of t-2-hex pro-apoptotic stimulation. To this end, we constructed a set of live cell reporters expressing destabilized luciferase (**Figure 3B** [↗](#)) under the control of transcription factor-specific response elements. We included binding sites for the Hsf1 and Rpn4 factors responding to protein aggregation and misfolding (HSE and PACE, respectively), the Pdr1/3 factors responding to xenobiotic stress (PDRE), and the Yap (Yap1 and other Yap TFs) and Sko1 factors responding to oxidative stress (AP-1 and CRE, respectively). We quantified the dose response of each reporter separately upon continuous increase of t-2-hex concentrations by time elapsed luciferase measurements. While all transcriptional reporters were robustly activated by the bioactive lipid, we found important differences in the sensitivity of the response for the different stress types. The Hsf1- and Rpn4-mediated transcriptional regulation showed the most sensitive activation upon t-2-hex stress with the lowest EC_{50} values of 8 and 10 μ M, respectively (**Figure 3C** [↗](#) and **D** [↗](#)). In contrast, the oxidation specific reporters were activated with significantly lower sensitivity. These results confirmed that the transcriptional heat shock and proteasomal responses were preferentially triggered by the pro-apoptotic lipid, presumably because it caused an important problem of protein misfolding in the cell.

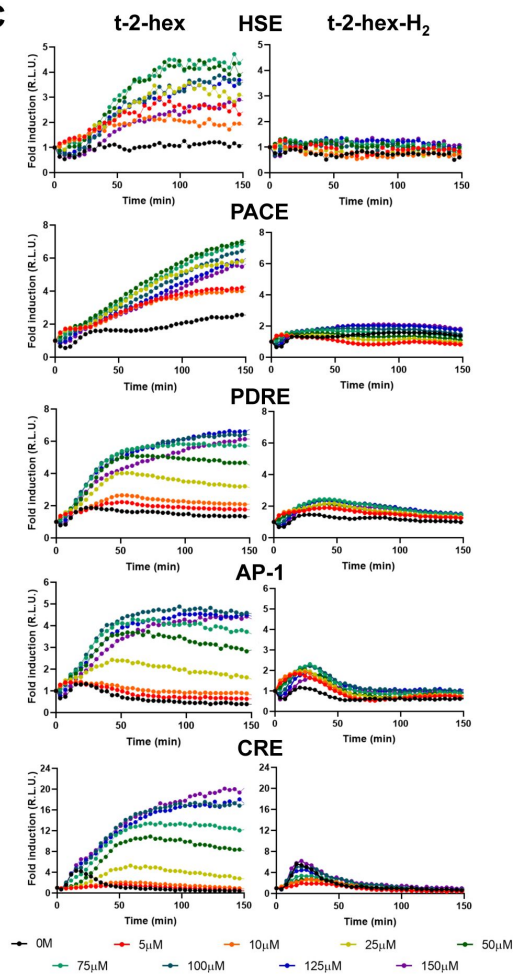
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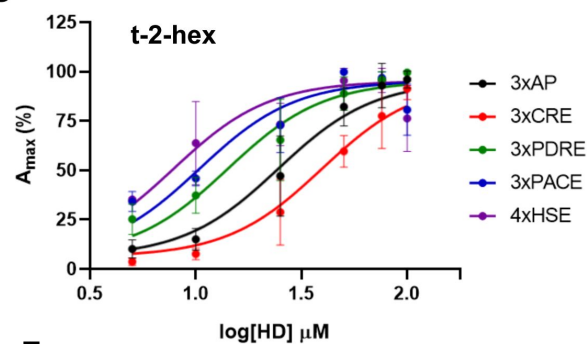
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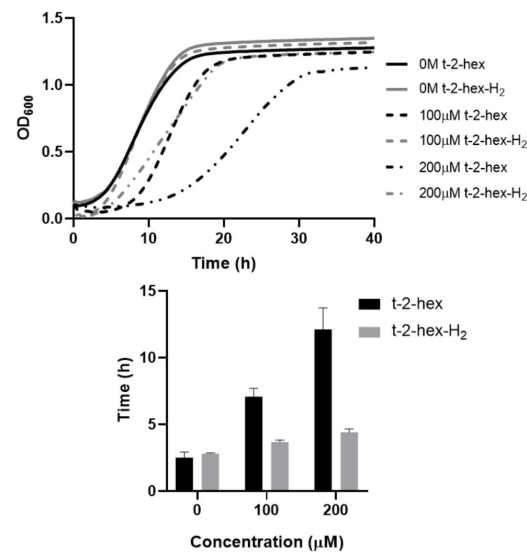


Figure 3.

t-2-hex specifically induces the heat shock and proteasomal transcriptional response. (A) Significantly enriched promoter motifs were identified from the transcriptionally up- and down-regulated genes identified by our RNA seq study upon t-2-hex stress. (B) Schematic representation of the stress-specific luciferase reporters applied in the dose-response experiments upon unsaturated t-2-hex or saturated analogue t-2-hex-H₂. The t-2-hex dose causing half maximal induction for each reporter is given in μ M below the constructs. (C) Dose-response curves of the indicated live cell luciferase reporters upon increasing t-2-hex and t-2-hex-H₂ concentrations (n = 3). All reporters were assayed in *hfd1Δ* cells. Initial light emission levels at time 0 were set to 1. (D) Comparison of the sensitivities of different stress type specific responses to the pro-apoptotic t-2-hex. Experimental data from (C) were analyzed by plotting the maximal reporter activation against the log[t-2-hex] concentration. (E) t-2-hex unsaturation is the cause of its severe growth inhibition. Growth of yeast wild type cells was scored upon the indicated t-2-hex and t-2-hex-H₂ concentrations (upper panel) and the lag time calculated (lower panel), n = 3.

Pro-apoptotic t-2-hex causes cytosolic protein aggregation

t-2-hex is an α/β unsaturated lipid aldehyde. In mammalian cells, t-2-hex exerts specific apoptotic functions by nucleophilic addition and direct lipidation of sulfhydryl groups of cysteines in the Bax protein (Cohen et al., 2020 [DOI](#)). We wanted to discern the biological effects of t-2-hex in yeast by comparing it to the saturated t-2-hex analogue hexadecanal or t-2-hex-H₂. It is specifically missing the ability to perform a nucleophilic addition and direct lipidation, therefore an absence of response upon exposure to the saturated compound indicates it is the lipidation that is responsible for the biological effects of t-2-hex. We found that the lack of α/β unsaturation in the lipid aldehyde changed the spectrum and intensity of stress responsive transcriptional activation according to our live cell luciferase reporter assays. t-2-hex-H₂ mainly triggered an antioxidant and xenobiotic stress response, while it did not activate the protein misfolding response through Hsf1 or Rpn4 (Figure 3B [DOI](#) and 3C [DOI](#)). Interestingly, t-2-hex-H₂ caused a typical transient activation of stress gene expression as compared to the sustained response triggered by t-2-hex. Accordingly, we found that t-2-hex-H₂ caused a significantly shorter growth inhibition as compared to the unsaturated t-2-hex (Figure 3E [DOI](#)). These data indicated that the pro-apoptotic t-2-hex induced a long lasting proteostatic emergency in the cell, possibly by direct lipidation of effector proteins. We next wanted to know whether t-2-hex was able to induce a general protein folding and aggregation problem in the cytosol. We monitored the intracellular distribution of the protein chaperone Hsp104, which was tagged with green fluorescent protein (GFP). Hsp104, together with Hsp70, is a molecular disaggregase (Glover and Lindquist, 1998 [DOI](#)) and can be used to visualize the cytosolic accumulation of protein aggregates (Erjavec et al., 2007 [DOI](#); Jacobson et al., 2017 [DOI](#)). The exposure to t-2-hex caused the appearance of multiple foci of protein aggregates in wild type cells, which increased in number over the first 6 hours of lipid stress (Figure 4A [DOI](#)). Larger protein aggregates were observed in cells with impaired proteasome function (*rpn4Δ*), indicating that proteasomal activity was needed to clear lipid-induced protein aggregates in the cytosol. Interestingly we observed that t-2-hex-H₂ only caused a modest and transient protein aggregation within the first hour of exposition (Figure 4A [DOI](#)). Thus, the massive protein aggregation, which mounts slowly over time, was specifically induced by the unsaturated t-2-hex molecule. Protein aggregation can occur at mature and/or newly synthesized proteins depending on the nature of the proteostatic stress. Accordingly, the application of the translation inhibitor cycloheximide revealed that Hsp104-foci are formed independently of protein synthesis in the case of heat stress, but their formation depends on protein synthesis in the case of arsenite stress (Jacobson et al., 2012 [DOI](#)). We found that cycloheximide pre-treatment of the cells completely abolished t-2-hex triggered protein aggregates (Figure 4B [DOI](#)), indicating that the pro-apoptotic lipid caused misfolding of *de novo* synthesized proteins.

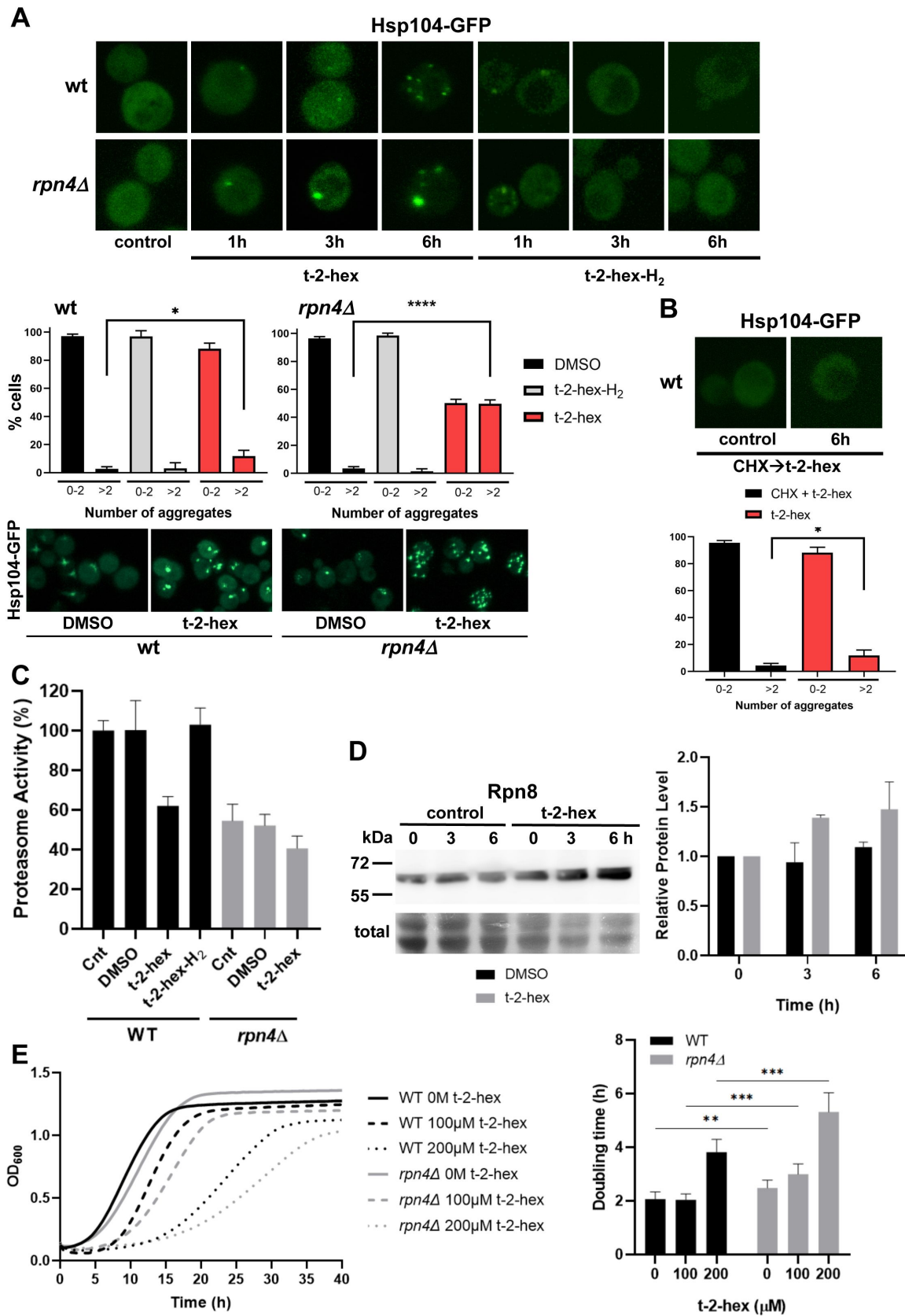


Figure 4.

t-2-hex leads to cytosolic protein aggregation and inhibition of proteasomal function. (A) GFP-tagged Hsp104 was used to visualize intracellular protein aggregation in wild type and proteasome deficient *rpn4Δ* cells. Specifically, unsaturated t-2-hex caused a slowly increasing protein aggregation. Cells were treated with 100 μM of the bioactive lipids for the indicated times. Lower panels: Quantification of t-2-hex induced protein aggregates across cell populations. Number of analyzed cells: wt DMSO n = 601, wt t-2-hex n = 584, wt t-2-hex-H₂ n = 558; *rpn4Δ* DMSO n = 551, *rpn4Δ* t-2-hex n = 549, *rpn4Δ* t-2-hex-H₂ n = 329. (B) t-2-hex activated protein aggregation was no longer observed after inhibition of protein synthesis with cycloheximide (CHX). Lower panel: Quantification of t-2-hex induced protein aggregates across cell populations. Number of analyzed cells: t-2-hex n = 584, CHX + t-2-hex n = 134. (C) Effect of pro-apoptotic t-2-hex on proteasomal activity. Proteasomal activity was quantified in whole cell extracts before (Cnt) or after treatment with 200 μM t-2-hex, t-2-hex-H₂ or vehicle (DMSO) for 3h in wild type or *rpn4Δ* cells (n = 3). Activity of untreated wild type cells was set to 100%. (D) Response of proteasomal subunit Rpn8 expression upon t-2-hex exposure. Rpn8 was expressed as a Tap fusion from its chromosomal locus and cells were treated or not with 200 μM t-2-hex for the indicated times. Rpn8 protein abundance was quantified by anti-Tap western blot (upper panel) and quantified relative to uninduced levels (n = 2). (E) Proteasomal deficiency causes t-2-hex sensitivity. Growth of yeast wild type and *rpn4Δ* cells was quantified upon the indicated t-2-hex concentrations (left panel) and the doubling time calculated (right panel), n = 6. *p<0.05, **p<0.01, ***p<0.001, ****p<0.0005 by Student's unpaired t-test.

The ubiquitin proteasome system is a main controller of cellular proteostasis (Dikic, 2017). Our transcriptomic data identified a uniform up-regulation of proteasomal subunits and activators upon t-2-hex stress. On the other hand, the proteasome cannot directly degrade aggregated proteins and instead is sensitively inhibited by aggregates (Bence et al., 2001; Tyedmers et al., 2010). To further explore the function of the proteasome during pro-apoptotic lipid stress, we determined the regulation of proteasomal activity. We measured the chymotrypsin-like activity of the proteasome in cell extracts using a synthetic substrate and found that t-2-hex, but not t-2-hex-H₂, severely diminished the activity of the proteasome in wild type yeast cells to levels comparable with the proteasome deficient *rpn4Δ* mutant (Figure 4C). We next quantified the protein levels of the proteasomal subunit Rpn8 and confirmed a continuous up-regulation upon exposure to t-2-hex (Figure 4D), thus confirming the RNA seq results. These data might be explained by an inhibition of the proteasomal activity by extensive protein aggregates formed upon t-2-hex stress, which the cell might want to counteract by stimulated proteasomal gene expression. We further confirmed that *rpn4Δ* cells were hypersensitive to pro-apoptotic t-2-hex treatment (Figure 4E).

Pro-apoptotic t-2-hex efficiently impairs mitochondrial protein import

Having established that pro-apoptotic lipid stress elicits a strong activation of mitochondrial protein import surveillance and general proteostatic rescue systems, we tested directly, whether t-2-hex impaired the import of mitochondrial precursor proteins. A defect in the import of mitochondrial proteins with a cleavable signal peptide can be visualized by the immunological detection of the larger, unprocessed pre-protein. We chose three different mitochondrial proteins for this survey: Aim17, whose expression is strongly activated upon t-2-hex stress (42 fold according our RNA Seq study), the Cox5a subunit of CIV at the inner mitochondrial membrane and the Ilv6 regulator of isoleucine and valine biosynthesis in the mitochondrial matrix. We found for all three proteins that t-2-hex treatment of wild type cells led to the rapid appearance of the corresponding pre-proteins, indicating an efficient block of mitochondrial protein import by the lipid (Figure 5A). The Aim17 and Cox5a pre-proteins correlated with the same precursors induced by the general mitochondrial uncoupler CCCP. However, in the Ilv6 case, we observed the appearance of yet an additional precursor form specific for t-2-hex. The behavior of Aim17 is especially interesting, because this protein is massively made *de novo* in response to t-2-hex exposure. We observe that its pre-protein quickly accumulates to very high levels, which equal the

abundance of the mature Aim17 protein. This strongly suggests that t-2-hex very efficiently blocks mitochondrial pre-protein transport and processing, and that most of the newly synthesized mitochondrial precursors are not correctly introduced into mitochondria upon pro-apoptotic lipid stress. We next confirmed that this inhibition was specific for the unsaturated t-2-hex form, as we did not detect precursor accumulation of four different mitochondrially imported proteins by the saturated t-2-hex-H₂ (**Figure 5B** [↗](#)). We additionally determined general mitochondrial fragmentation, which had been reported previously for t-2-hex (Manzanares-Estreder et al., 2017 [↗](#)). Also in this case, we found that mitochondrial inhibition was specific for the unsaturated lipid aldehyde (**Figure 5C** [↗](#)), indicating that mitochondrial fragmentation and inhibition of protein import depended on the protein lipidation capacity of t-2-hex.

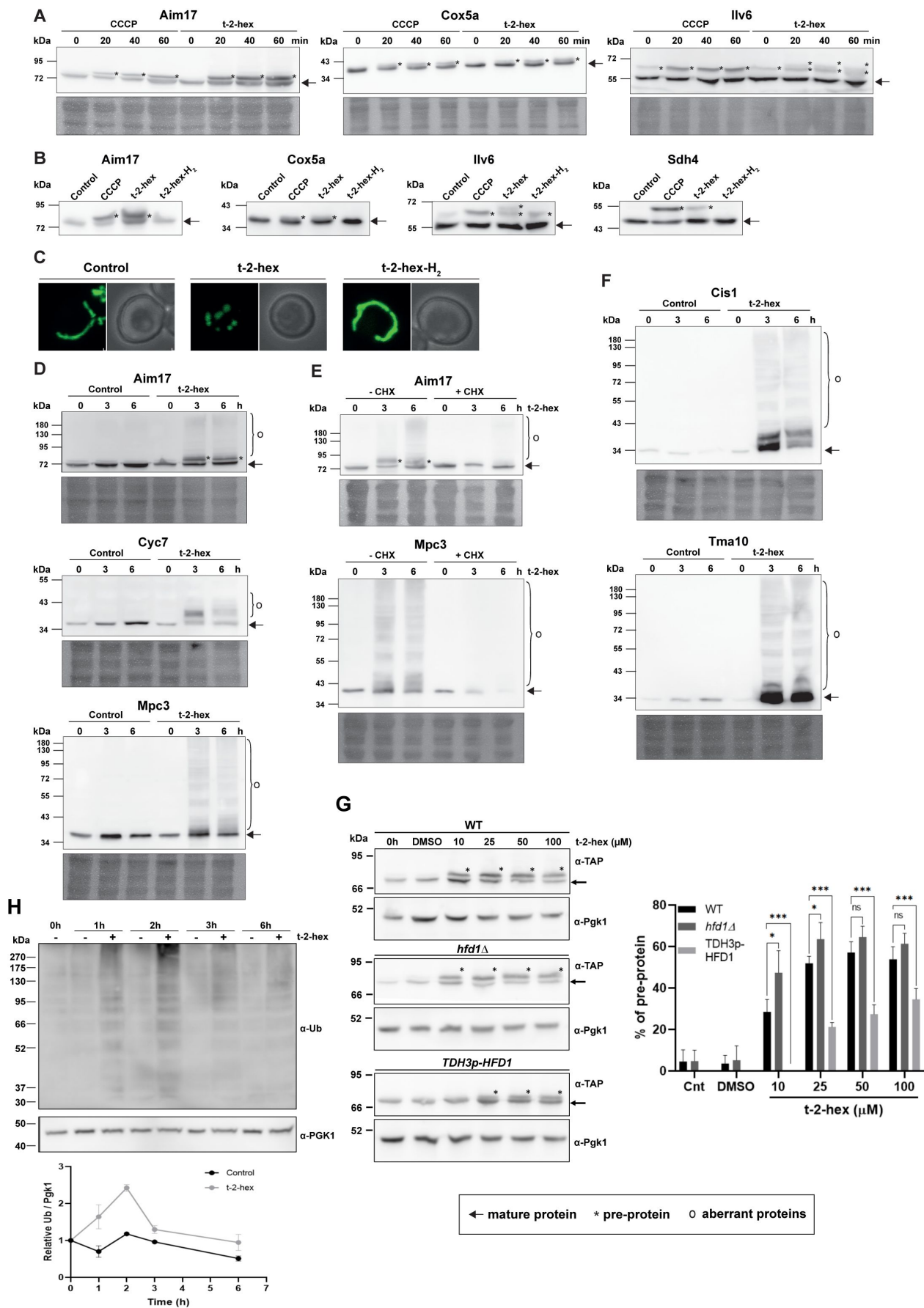


Figure 5.

t-2-hex overload induces mitochondrial pre-protein accumulation and aggregation of *de novo* synthesized proteins. (A) The appearance of unimported mitochondrial precursor proteins (*) was induced by the uncoupler CCCP (20μM) or t-2-hex (200μM) for the indicated times. Aim17, Cox5a, Ilv6 and Sdh4 were visualized in chromosomally Tap-tagged wild type yeast strains by anti-Tap western blot. (B) Mitochondrial import block depends on t-2-hex unsaturation. The same strains as in (A) were treated with DMSO (control), CCCP (20μM), t-2-hex (200μM) or t-2-hex-H₂ (200μM) for 40min and mitochondrial pre-protein accumulation visualized by anti-Tap western blot. (C) Mitochondrial fragmentation depends on t-2-hex unsaturation. Yeast wild type cells expressing mt-GFP were treated with vehicle (control), 200μM t-2-hex or t-2-hex-H₂ for 1h. (D) t-2-hex induces the formation of aberrant mitochondrial proteins. Yeast wild type strains expressing Aim17-, Cyc7- or Mpc3-Tap tagged fusion proteins from their chromosomal locus were treated with 200μM t-2-hex for the indicated times. Fusion proteins were detected by anti-Tap western blot. (E) Inhibition of *de novo* protein synthesis abolishes the formation of aberrant mitochondrial proteins. Experimental conditions as in (D), but including a pretreatment with cycloheximide (CHX, 250μg/ml) where indicated. (F) t-2-hex induces aberrant forms of highly expressed proteins. Yeast wild type strains expressing Cis1- or Tma10-Tap tagged fusion proteins from their chromosomal locus were treated or not with 200μM t-2-hex for the indicated times. Fusion proteins were detected by anti-Tap western blot. (G) Hfd1 gene dose determines the extent of t-2-hex induced pre-protein accumulation. Aim17 pre-protein accumulation in response to different t-2-hex concentrations was quantified in wild type, *hfd1Δ* and *HFD1* overexpressing cells. Cells were either untreated, or treated with the indicated lipid concentrations or DMSO for 20 min. The percentage of pre-protein relative to total Aim17 protein was calculated in the right panel (n = 4) (H) t-2-hex causes rapid protein ubiquitination. Yeast wild type cells were treated for the indicated times with t-2-hex (200μM) and protein ubiquitination visualized by anti-Ub western blot (upper panel) and quantified relative to the Pgk1 loading control (lower panel) (n = 2).

The observation that newly made Aim17 was not processed properly and accumulated in large amounts during the first hour of t-2-hex stress, prompted us to investigate the long-term effects of t-2-hex on proteostasis. For these studies, we initially selected two more highly inducible mitochondrial proteins apart from Aim17, the cytochrome c variant Cyc7 targeted to the mitochondrial intermembrane space and the pyruvate carrier subunit Mpc3 targeted to the inner mitochondrial membrane. All proteins were visualized in yeast wild type cells, which were treated with t-2-hex for up to 6 hours. As expected, we found that the Aim17 pre-protein also accumulated at prolonged t-2-hex exposure (**Figure 5D**). However, we additionally were able to see a smear of aberrant, SDS resistant protein forms in the larger molecular weight range, which could be an indication of aggregated Aim17. Aberrant protein forms were also detected for Cyc7 and, in a very abundant manner, for Mpc3, which both lack a defined cleavable signal peptide. Although the expression of all three mitochondrial proteins was activated, the amount of the mature protein forms did not increase (**Figure 5D**). This suggested that the newly synthesized mitochondrial proteins never get properly incorporated in the organelle upon t-2-hex stress and instead remain in the cytoplasm, where they increasingly misfold and aggregate. Indeed, cycloheximide treatment completely abolished the formation of Mpc3 and Aim17 aggregates (**Figure 5E**), indicating that t-2-hex only affects the proteostasis of *de novo* synthesized mitochondrial precursor proteins.

Clogging the mitochondrial import machinery has been shown to cause a generalized proteostatic problem in the cytoplasm, most likely by saturating the cellular chaperone system (Boos et al., 2019; Wrobel et al., 2015). We therefore looked at other t-2-hex inducible proteins, which are not transported through the mitochondrial membranes, Cis1 and Tma10. The MitoCPR protein Cis1 is induced upon mitochondrial protein import stress and attaches to the outer mitochondrial transport complex without being imported. Tma10 is the most up-regulated protein upon t-2-hex stress according to our RNA-Seq analysis and localizes to the cytoplasm. For both proteins, we confirm a strong increase in abundance upon t-2-hex treatment (**Figure 5F**). More importantly, also Cis1 and Tma10 show an abundant aggregation when they are massively synthesized in the presence of the pro-apoptotic lipid (**Figure 5F**), suggesting that t-2-hex probably affects correct

folding of a large number of newly synthesized mitochondrial and cytosolic proteins. We next tested how sensitively t-2-hex interfered with mitochondrial protein import by applying low lipid doses. As shown in **Figure 5G**, Aim17 pre-protein accumulated by the external addition of just 10 μ M of t-2-hex. This inhibition was aggravated in a *hfd1 Δ* mutant and alleviated by *HFD1* overexpression (**Figure 5G**). These data indicated that mitochondrial pre-protein accumulation was induced by physiological t-2-hex concentrations and that its extent was regulated by the activity of detoxifying enzymes such as Hfd1. Misfolded cytosolic proteins are rapidly marked by ubiquitination for their proteolytic degradation by the proteasome. We therefore tested whether t-2-hex overload increased overall protein ubiquitination. We performed anti-Ub western blot analyses in a time course of t-2-hex treatment in whole cell extracts. As shown in **Figure 5H**, t-2-hex induced general protein ubiquitination in a rapid and transient manner.

A functional genomics approach reveals pro- and anti-apoptotic gene functions in response to t-2-hex

In order to identify genetic determinants, which would modulate pro-apoptotic t-2-hex toxicity, we performed a saturated transposon mutagenesis screen (SATAY; (Michel et al., 2017)). We generated a yeast mutant library consisting of several millions of independent transposon insertion mutants in an *hfd1 Δ* background (**Figure 6A**). The pooled library was then grown in the presence or absence of growth-limiting amounts of t-2-hex. The transposon insertion sites (TNs) were mapped along the entire yeast genome by massive sequencing. The total number of transposon reads or the number of different TNs per gene was then used as an indicator of the fitness of all mutants upon pro-apoptotic lipid stress conditions. In this scenario, under-represented genes (statistically lower number of TNs in a given gene in the t-2-hex versus the untreated library) have potentially anti-apoptotic functions against the bioactive lipid (Supplemental Table 4), while over-represented genes (statistically higher number of TNs in a given gene) have potentially pro-apoptotic functions supporting the toxicity of t-2-hex (Supplemental Table 5). We analyzed the potential anti-apoptotic SATAY hits with a $\log_2 < -0.75$ according to expected detoxification pathways (heat shock response, pleiotropic drug response), to their function in the ER (the intracellular site where t-2-hex is generated) or in mitochondria (the major t-2-hex target identified so far). As depicted in **Figure 6B**, transposon mutations in the *HSF1* gene encoding heat shock factor are strongly counter-selected by t-2-hex stress. *HSF1* is essential in yeast, however, mutations in the Hsf1 C-terminal region are viable and reduce the activity of the transcription factor (Chen et al., 1993). Untreated cells accumulate TN insertions in this region, however, the Hsf1 C-terminus becomes essential upon t-2-hex treatment, thus demonstrating that the heat shock response is rate limiting during the cellular defense against the lipid. Additionally, the pleiotropic drug response (PDR) is identified here as an important layer of defense against t-2-hex. Both the dominant PDR transcriptional activator Pdr1 and the Snq2 PDR extrusion pump of the ABC type are among the gene functions, which are not allowed to accumulate TN insertions upon t-2-hex overload. Furthermore, we identify several mitochondrial functions (*FMP52*, *CYC3*, *HEM14*, *MRP7*), which become important upon apoptotic lipid stress. They play essential functions in heme biosynthesis (Hem14) and cytochrome c maturation (Cyc3), or in mitochondrial protein synthesis (Mrp7). Fmp52, however, is a mitochondrial protein with unknown function. Of note, Fmp52 is a member of the short-chain dehydrogenase/reductase (SDR) family, which suggests a function in the enzymatic detoxification of t-2-hex additionally to Hfd1. We furthermore identify a group of ER-resident proteins as crucial for t-2-hex tolerance. Apart from the ER membrane proteins Agp2, Nnf1, Cwh43 and Spf1, we found two proteins, Eps1 and Usa1, specifically involved in the ER associated protein degradation pathway (ERAD) as important for t-2-hex tolerance.

Unexpectedly, the SATAY screen revealed a large number of genes, which upon TN mutation, favored proliferation upon pro-apoptotic t-2-hex stress (**Figure 6B**). A closer inspection of these gene functions with a $\log_2 > 1.5$ revealed that most of them belonged to the functional groups “Cytosolic ribosome and translation” and “Amino acid metabolism”. Related to cytosolic protein

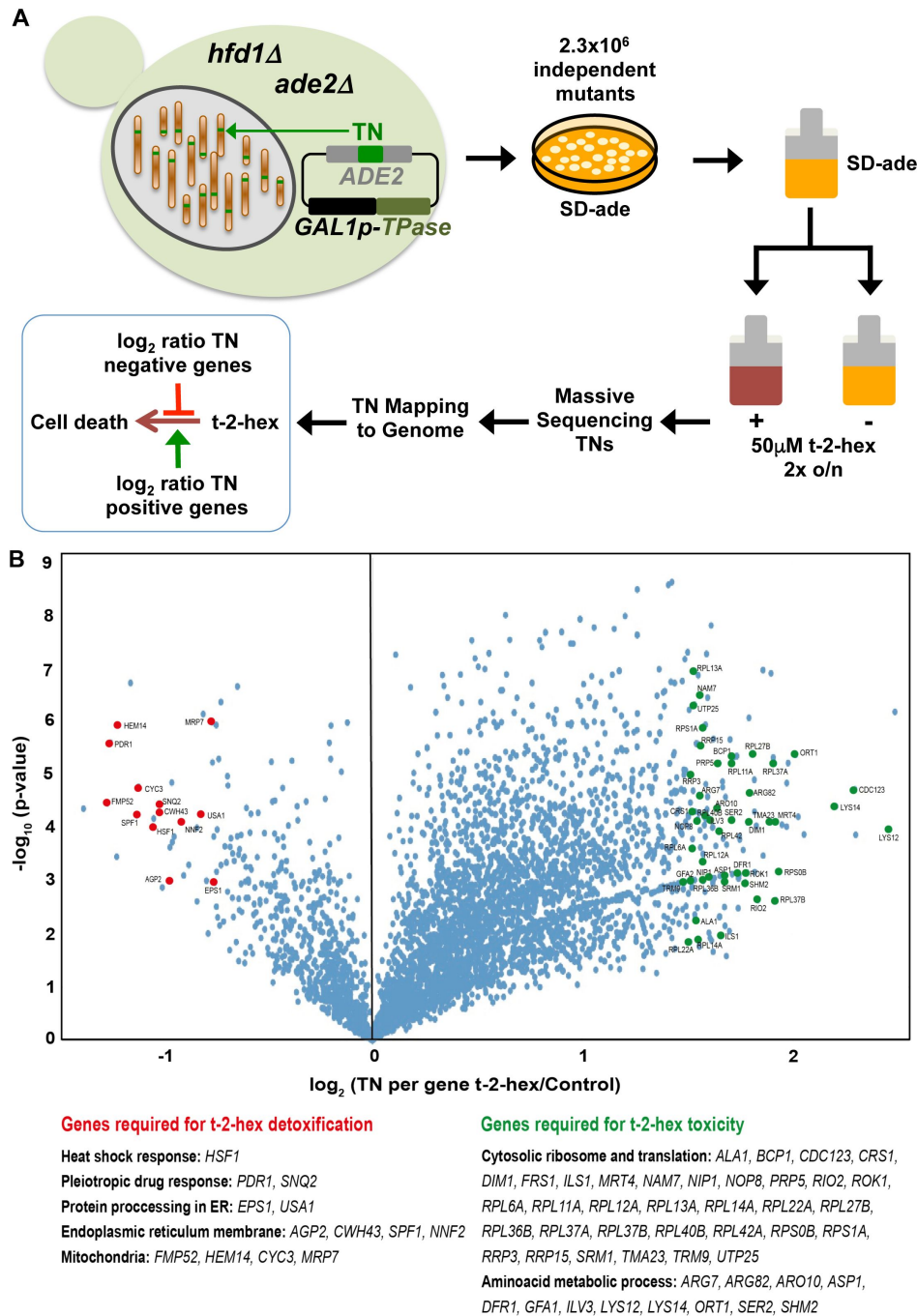


Figure 6.

Functional genomics screen SATAY for the identification of pro- and anti-apoptotic functions upon t-2-hex stress. (A) Schematic outline of the SATAY experiment. *hfd1Δ* cells harboring the galactose inducible transposase (TPase) and a transposon (TN) disrupting the *ADE2* gene were grown in galactose-containing medium to induce transposition. TN generated mutant cells were inoculated in synthetic glucose medium lacking adenine in the presence or absence of 50μM t-2-hex, grown for several generations and harvested for DNA extraction and sequencing of transposon insertion sites (TNs). TNs are mapped to the genome to identify genes that become required for proliferation under t-2-hex stress conditions (red) or genes whose mutation is beneficial for t-2-hex tolerance (green). (B) Identification of anti- and pro-apoptotic gene functions upon t-2-hex overload. Volcano plot showing the fold change of number of transposon insertions (TN) per gene of libraries grown in t-2-hex excess versus control conditions. TN under-enriched (anti-apoptotic) genes were analyzed with a log₂ ratio below -0.75 and are summarized in Supplemental Table 3. TN enriched (pro-apoptotic) genes were analyzed with a log₂ ratio > 1.5 and are available in Supplemental Table 4.

biosynthesis, we identified proteins involved in ribosomal RNA processing (*DIM1*, *RIO2*, *ROK1*, *RRP3*, *RRP15*, *SRM1*), ribosome biogenesis (*MRT4*, *NOP8*, *BCP1*, *TMA23*, *UTP25*), aa-tRNA synthesis (*ALA1*, *CRS1*, *FRS1*, *ILS1*, *TRM9*), pre-RNA processing (*PRP5*), translation (*CDC123*, *NIP1*, *NAM7*), and ribosomal protein subunits (*RPL6A*, *RPS1A* + 12 others). Related to amino acid metabolism, enzymes providing precursors for general amino acid biosynthesis (*ARO10*, *DFR1*, *SHM2*), as well as regulators and enzymes of the synthesis of specific amino acids (*ARG7*, *ARG82*, *ASP1*, *GFA1*, *ILV3*, *LYS12*, *LYS14*, *ORT1*, *SER2*) were identified in this group of potential pro-apoptotic functions. Taken together, these results suggested that an efficient heat shock and PDR response, as well as specific mitochondrial and ER functions were required, while a high rate of cytosolic protein biogenesis was detrimental, for the survival upon apoptotic lipid stress.

Indeed, we confirmed that loss of function of the PDR transcriptional activator Pdr1 caused a moderate, but consistent growth delay in the presence of t-2-hex (**Figure 7A** [↗](#)). Interestingly, we found that deletion of mitochondrial Fmp52 function caused a severe sensitivity to t-2-hex exposure (**Figure 7A** [↗](#)). Fmp52 belongs structurally to the NAD-binding short-chain dehydrogenase/reductase (SDR) family and localizes to highly purified mitochondrial outer membranes (Zahedi et al., 2006 [↗](#)). This might suggest that Fmp52 participates in the enzymatic detoxification of t-2-hex in addition to Hfd1.

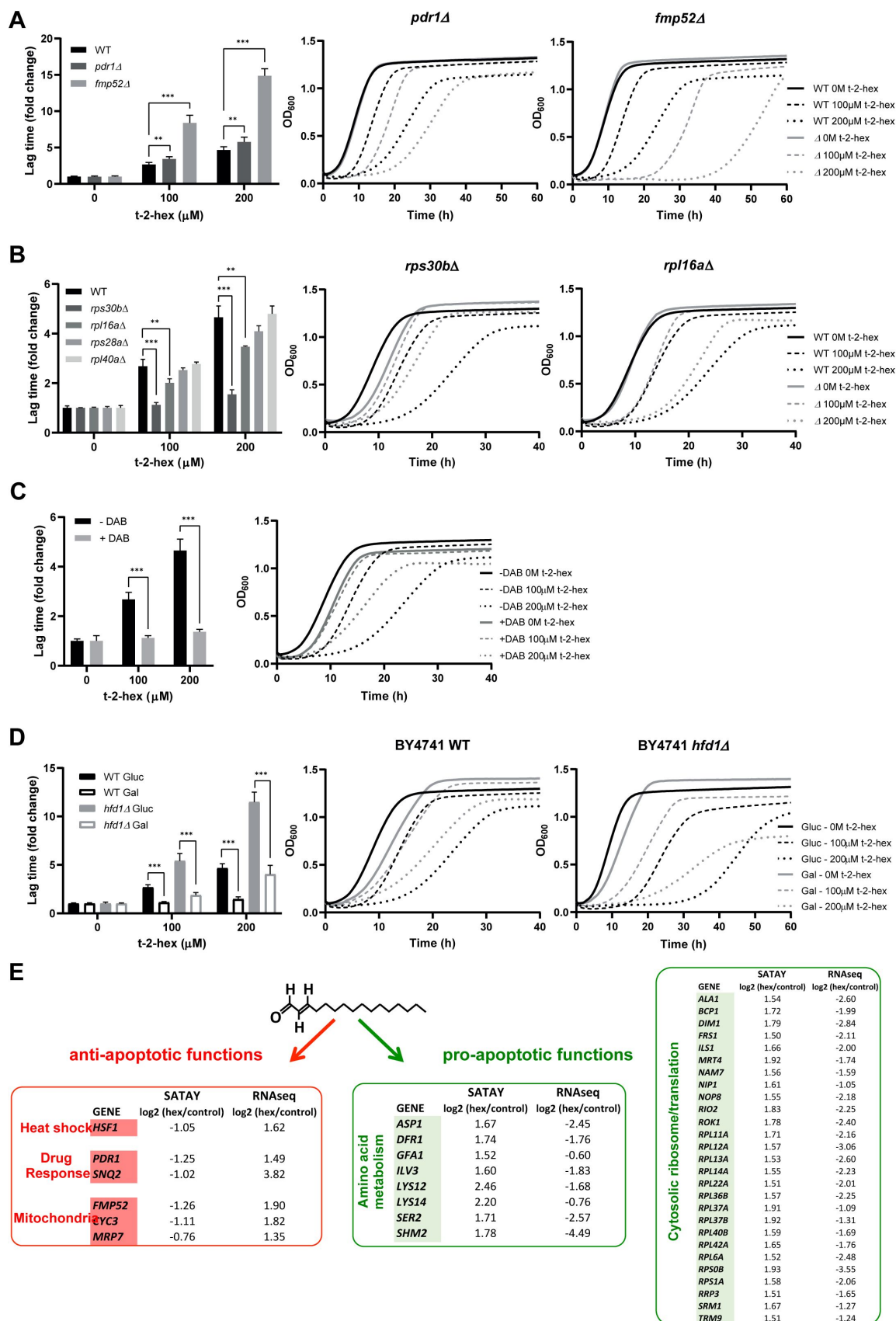


Figure 7.

Functional analysis of pro- and anti-apoptotic genes upon t-2-hex excess. (A) The pleiotropic drug response activator Pdr1 and the mitochondrial protein Fmp52 are necessary for t-2-hex tolerance. Quantitative growth assay of wild type, *pdr1Δ* and *fmp52Δ* cells upon the indicated concentrations of t-2-hex. The relative lag time was quantified as an estimator of growth inhibition (n=3). (B) Deletion of ribosomal protein subunits causes t-2-hex resistance. As in (A), but using the yeast deletion strains *rps30bΔ*, *rpl6aΔ*, *rps28aΔ* and *rpl40aΔ*. Growth curves are depicted only for the t-2-hex tolerant strains *rps30bΔ* and *rpl6aΔ* (n=3). (C) Inhibition of ribosomal biogenesis is beneficial for t-2-hex tolerance. Yeast wild type cells were grown upon the indicated t-2-hex concentrations in the presence or absence of the inhibitor diazaborine (DAB, 20μg/ml) (n=3). (D) Galactose growth improves t-2-hex tolerance. Growth of wild type and *hfd1Δ* cells upon the indicated concentrations of t-2-hex on synthetic glucose or galactose media (n=3). **p<0.01, ***p<0.001 by Student's unpaired t-test. (E) Summary of anti-apoptotic genes identified by SATAY with significant transcriptional up-regulation (red) and pro-apoptotic genes identified by SATAY with significant transcriptional down-regulation (green) according to our RNA-seq study.

We next wanted to know whether a general connection existed between the *de novo* protein synthesis rate and the susceptibility to pro-apoptotic t-2-hex. As a first approach, we tested several deletion mutants in specific subunits of the small and large ribosomal subunits from the yeast knock out collection for their individual t-2-hex tolerance. As it is known that ribosomal subunit mutants have divergent effects on protein synthesis rates and rapidly accumulate compensatory mutations (Steffen et al., 2008 [↗](#)), we included several mutant strains with diverse impact on overall growth rates in this study. As shown in **Figure 7B** [↗](#), the *rps30b* mutant had the greatest reduction in proliferation rate and showed the highest tolerance to t-2-hex. The *rpl16a* mutant with a minimal growth reduction was only slightly resistant to the apoptotic lipid, while other ribosomal subunit mutants (*rps28a*, *rpl40a*) with unaffected growth rates were t-2-hex sensitive comparable to the wild type. This indicated that the loss of particular ribosomal subunits was able to improve t-2-hex tolerance in cases where growth rate and presumably protein synthesis rates were decreased. We then tested this hypothesis directly by the application of the ribosomal inhibitor diazaborine (DAB), which specifically interferes with ribosomal biogenesis and protein synthesis in yeast (Pertschy et al., 2004 [↗](#)). As shown in **Figure 7C** [↗](#), growth inhibitory DAB concentrations consistently made yeast cells more tolerant to t-2-hex, indicating that a slow cytosolic protein synthesis was beneficial to counteract the apoptotic lipid. Finally, we applied an independent intervention to slow down protein synthesis, which is the switch from glucose to galactose growth. Metabolic flux through the galactose pathway is reduced by >30% as compared to glucose, which slows down cell proliferation (Ostergaard et al., 2000 [↗](#)). We confirmed that galactose growth improved the resistance of yeast cells to t-2-hex inhibition, both in the wild type and the *hfd1Δ* background (**Figure 7D** [↗](#)).

We next compared the pro- and anti-apoptotic gene functions indicated by the SATAY assay with the previously generated RNAseq profiles. We found that anti-apoptotic functions such as heat shock factor, pleiotropic drug response and specific mitochondrial proteins showed consistently positive transcriptional regulation (**Figure 7E** [↗](#), red), while the expression of the majority of pro-apoptotic functions involved in cytosolic protein synthesis and general amino acid metabolism was down-regulated (**Figure 7E** [↗](#), green).

Apoptotic lipid signaling via t-2-hex and its detoxification take place at the mitochondrial TOM complex

Having found that the pro-apoptotic lipid t-2-hex impaired mitochondrial protein import and caused a general proteostatic readjustment in the cell, we next wanted to know whether t-2-hex and its detoxifier Hfd1 target the mitochondrial pre-protein import machinery. The essential import pore at the outer mitochondrial membrane is formed by the TOM complex. The yeast core

TOM complex contains the Tom40, Tom22, Tom5, Tom6 and Tom7 subunits, which associate with the Tom70 and Tom20 receptor proteins to form the functional translocase complex (Araiso et al., 2022 [DOI](#)). We first focused on the interaction of Hfd1 with the TOM complex. It had been previously shown that the Hfd1 lipid aldehyde dehydrogenase co-purified with Tom22 (Opaliński et al., 2018 [DOI](#)) and Tom70 (Burri et al., 2006 [DOI](#)). Additionally Hfd1 seems to co-localize with a Tom70 containing complex in a recent high-resolution complexome profiling study (Schulte et al., 2023 [DOI](#)). We therefore studied the interaction of Hfd1 with Tom22 and Tom70 in targeted pulldown experiments from whole cell extracts. As shown in **Figure 8A** [DOI](#), we found that Hfd1 co-purified weakly with Tom22, however, it was more efficiently pulled down with Tom70. These data suggested that at least a portion of Hfd1 is physically associated with the TOM mitochondrial protein import machine with Tom70 as a potential interactor. These data indicated that t-2-hex might act directly at the TOM complex. We therefore proceeded with biochemical assays to elucidate a possible direct modification of TOM subunits by the apoptotic lipid.

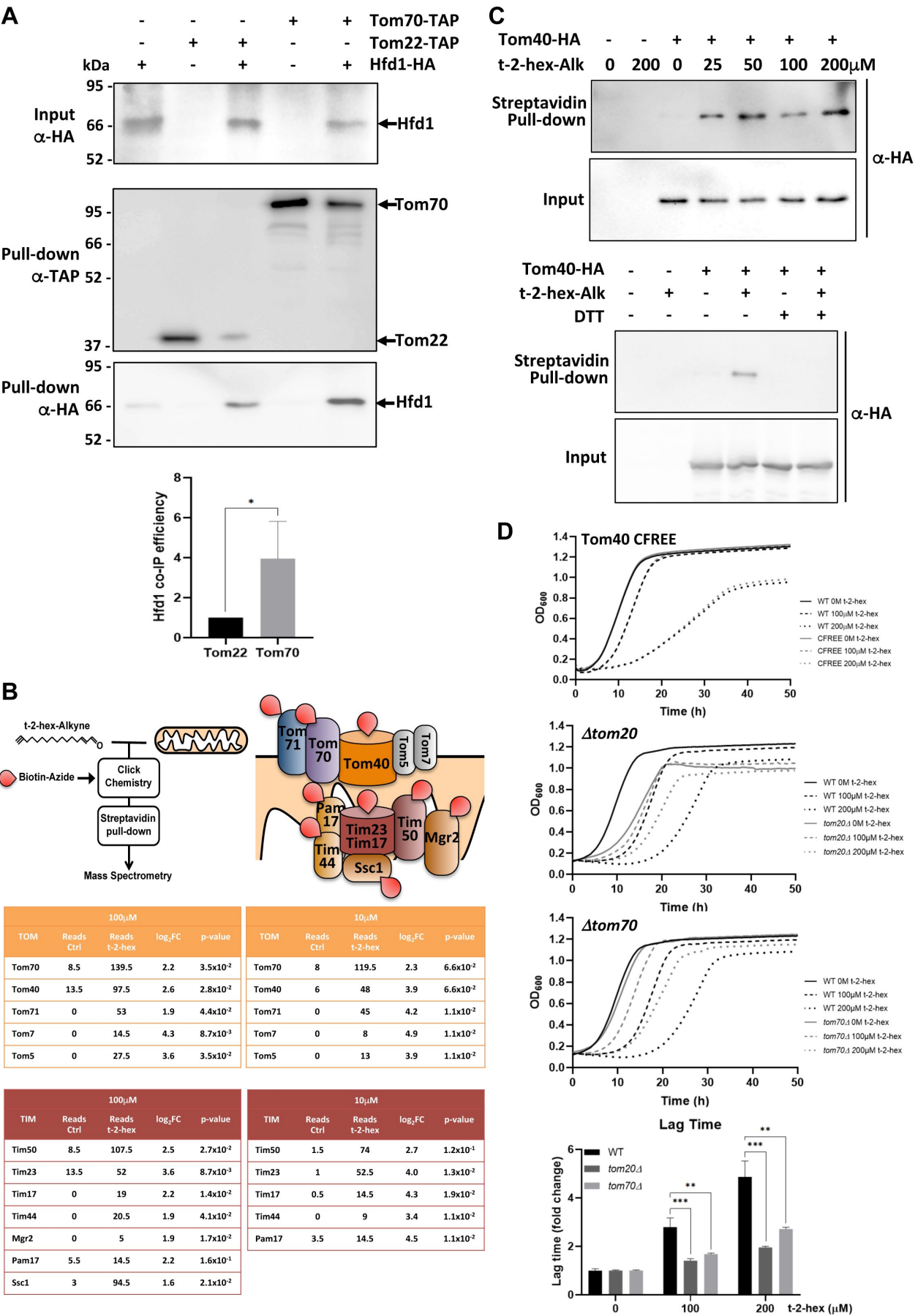


Figure 8.

t-2-hex targets the mitochondrial TOM complex. (A) Hfd1 co-purifies with Tom70. Constitutively Hfd1-HA expressing cells were used in co-immunoprecipitation experiments from yeast whole cell extracts in the presence or not of endogenously expressed Tom22- or Tom70-Tap. Lower panel: Quantification of Hfd1 co-purification with respect to Tom20 and Tom70; $n = 6$. (B) A chemoproteomic screen with t-2-hex alkyne identifies the TOM and Tim23 complexes as lipidation targets. Purified mitochondrial fractions from yeast wild type cells were treated or not with the clickable analogue t-2-hex-Alkyne at high and low doses (100 μ M or 10 μ M). After addition of biotin to the modified proteins and purification with Streptavidin agarose pull-down, the protein identities of the t-2-hex targets were determined by mass spectrometry. Cysteine containing subunits of the TOM and Tim23 complexes are depicted as direct lipidation targets. The tables show the mean spectral reads for selected subunits in the chemoproteomic analysis for mock treated (Ctrl) and t-2-hex treated mitochondria ($n = 2$), as well as the $\log_2$ fold enrichment and adjusted p-value. (C) Tom40 is lipidated *in vitro* by t-2-hex. Upper panel: t-2-hex-Alkyne was used in the indicated concentrations to lipidate proteins in enriched mitochondrial preparations from Tom40-HA expressing cells or control cells. After t-2-hex addition, input samples were generated directly, while pull-down samples were treated with click chemistry for covalent biotin linkage and subsequent Streptavidin purification. Tom40-HA was detected in all samples by anti-HA western blot. Lower panel: Competition of Tom40 t-2-hex lipidation by free thiol groups. t-2-hex-Alkyne was used at 100 μ M to lipidate Tom40-HA in mitochondrial preparations in the presence or absence of 2mM DTT. (D) Mutations of TOM accessory subunits 20 or 70 cause t-2-hex tolerance. t-2-hex toxicity assays comparing wild type cells with the cysteine free Tom40 mutant (Tom40 CFREE) and deletion mutants *tom20* or *tom70*. Lower panel: Quantitative comparison of growth performance (length of lag phase) of the *tom20* and *tom70* strains upon the indicated t-2-hex concentrations ($n = 3$). ** $p < 0.01$, *** $p < 0.001$ by Student's unpaired t-test.

We performed *in vitro* chemoproteomic experiments using the clickable t-2-hex analogue t-2-hex-alkyne on enriched intact mitochondrial fractions of yeast wild type cells, applying both high (100 μ M) and low (10 μ M) lipid doses (**Figure 8B**). After alkylation of mitochondrial proteins and the subsequent addition of biotin, we identified the t-2-hex targets by mass spectrometry. We found that several subunits of the TOM complex, including the Tom40 core channel and the accessory Tom70 and Tom71 proteins, were significantly enriched in the t-2-hex treated samples (**Figure 8B**, Supplemental Table 6). We additionally identified with less intensity the small Tom5 and Tom7 subunits, which lack cysteine residues and thus are most likely pulled down in these assays because of their physical association with truly lipidated Tom subunits. Very interestingly, we additionally identified several essential subunits of the Tim23 inner mitochondrial membrane protein import complex as t-2-hex targets (**Figure 8B**). This indicates that apart from the TOM complex, the pro-apoptotic lipid directly modifies subunits of the Tim23 protein translocator. Of the core TOM complex from yeast, only the Tom40 channel contains cysteine residues and interestingly has been identified as a t-2-hex target in a chemoproteomic screen on whole human cell extracts (Jarugumilli et al., 2018). We next tested for direct lipidation of Tom40 in mitochondrial fractions of yeast cells with physiological levels of HA-tagged Tom40. Low micromolar concentrations (25 μ M) of the t-2-hex probe were sufficient to specifically modify Tom40 in these lipidation assays (**Figure 8C**), showing that the Tom40 protein is a direct lipidation target *in vitro*. Competition assays with free thiol groups confirmed that t-2-hex likely adds to cysteine residues in the Tom40 channel (**Figure 8C**). We next addressed the question whether individual TOM subunits were important to mediate t-2-hex toxicity. We first characterized a mutant, which expressed a cysteine-free version of Tom40 (Tom40CFREE), thus impairing cysteine lipidation at Tom40. As shown in **Figure 8D**, the Tom40CFREE strain showed undistinguishable tolerance to t-2-hex as compared to wild type, suggesting that Tom40 lipidation alone did not explain the growth arrest of the pro-apoptotic lipid. We therefore extended our study to other, non-essential accessory TOM subunits Tom20 and Tom70. In this case (**Fig. 8D**), we observed a significant increase in t-2-hex tolerance caused by the deletion of either *TOM20* or *TOM70*. We further confirmed that a *tom20* mutant strain accumulated less Aim17 pre-protein upon t-2-hex stress (Supplemental Figure 1). These results identify the mitochondrial protein

import machinery as a physiologically important target for the pro-apoptotic lipid t-2-hex, which is mediated by several Tom subunits and not just the Tom40 import channel. In context with all previous results of our study, we propose a model where pro-apoptotic lipid signaling and cell proteostasis are functionally linked via mitochondrial protein import (**Figure 9**). Upon excess of t-2-hex or lack of Hfd1 function, lipidation of the TOM complex and subsequent failure of mitochondrial precursor import causes a general proteostatic imbalance. If this imbalance persists it can engage the cell in apoptotic cell death.

Discussion

Maintaining proteostasis is essential for cell survival and is tightly regulated to prevent the accumulation of misfolded or aggregated proteins. Proteostatic imbalance is known to be a key driver for the induction of apoptotic cell death and cellular aging (Kaushik and Cuervo, 2015). Although it has been well established that many extracellular stresses trigger the entry into apoptotic self-destruction programs and many executors of these apoptotic programs have been revealed (Kist and Vucic, 2021), it remains largely unknown how intracellular signals perturb mitochondrial function and proteostasis in order to push the balance towards a regulated cell death. Our present work reveals a novel pro-apoptotic mechanism, which consists of the direct inhibition of mitochondrial protein import by bioactive lipids generated from the sphingolipid degradation pathway, thereby directly linking pro-apoptotic lipids with mitochondrial and general protein homeostasis and aggregation.

It was a groundbreaking discovery that bioactive lipids such as t-2-hex in ER microsomes were in great part necessary and responsible for MOMP and mitochondrial cell death in human cells (Chipuk et al., 2012, 2021; Renault and Chipuk, 2013). The recent finding that t-2-hex is a direct pro-apoptotic modulator of a single cysteine residue in human Bax provided an explicit molecular mechanism of MOMP induction (Cohen et al., 2020; Jarugumilli et al., 2018). However, t-2-hex belongs to a group of lipid-derived electrophiles (LDEs), which are capable of directly modifying free cysteines in target proteins in a non-enzymatic manner and have been suggested to act as endogenous messengers to modulate signal transduction pathways within the cell (Fritz and Petersen, 2013; Sousa et al., 2017). This implies that t-2-hex has potentially many protein targets in the cell, which has been further demonstrated by *in vitro* chemoproteomic experiments in human cells, revealing several hundred alkylation targets including Tom40 (Jarugumilli et al., 2018). It remains challenging and important to identify the biologically relevant protein targets of LDEs and especially t-2-hex (Chen et al., 2016; Wang et al., 2014). Our combination of *in vitro* lipidation assays and functional genomics surveys identifies the mitochondrial outer membrane protein transport machine as a selective target for apoptotic signaling mediated through t-2-hex. Moreover, the lipid-mediated inhibition of the mitochondrial TOM complex, here detailed in the yeast model, provides a general apoptotic mechanism, which might have been established early in evolution, given the conservation of the core subunits Tom40 or Tom70 from fungi to man (Mani et al., 2016).

Mitochondrial protein import might be a strategic target for apoptotic signaling, because the import of more than thousand cytoplasmatically synthesized pre-proteins into mitochondria is a major traffic hub in eukaryotic cells, which is under exquisite surveillance to avoid clogging of the organelle (Song et al., 2021). The aberrant accumulation of mitochondrial precursor proteins in the cytosol leads to proteotoxic stress, which negatively affects cell viability (Nargund et al., 2012; Wang and Chen, 2015; Wrobel et al., 2015). This phenomenon has been studied by the use of specific import mutants or the overexpression of artificial clogger proteins to induce a specific protein traffic jam at mitochondria. Indeed, the chronic impairment of mitochondrial protein import leads to cell death (Boos et al., 2019; Coyne et al., 2023; Wang and Chen, 2015; Wrobel et al., 2015). Cells have developed a coordinated and multifaceted response network to counteract the accumulation of these precursor proteins. This response network is

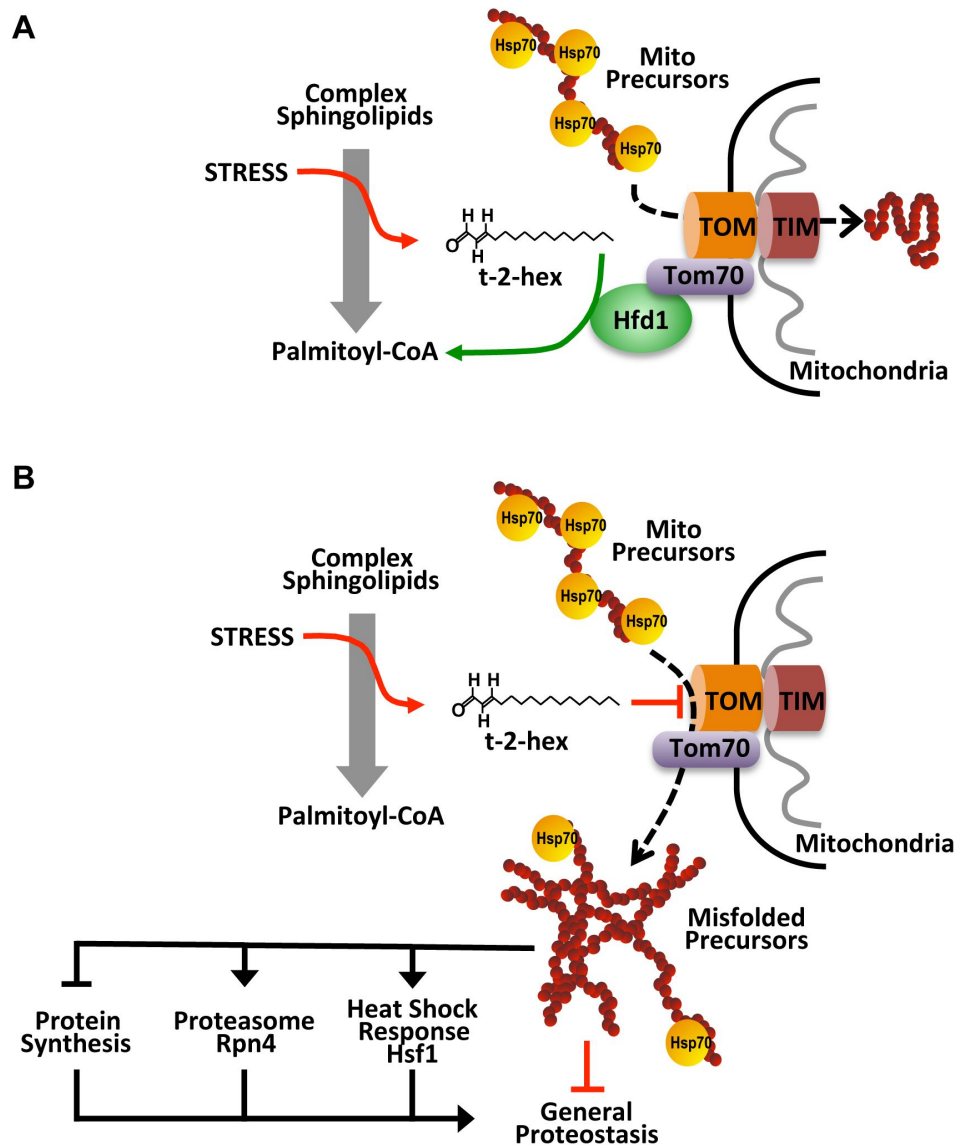


Figure 9.

Model of the pro-apoptotic function of t-2-hex and the anti-apoptotic function of Hfd1 at the mitochondrial Tom complex. Upper panel: Under normal conditions, the Hfd1 lipid aldehyde dehydrogenase located at the Tom complex safeguards mitochondrial protein import by t-2-hex degradation. Lower panel: Upon severe stress conditions or in the absence of Hfd1 function, an excess of t-2-hex directly lipidates Tom subunits such as Tom40 and thus inhibits mitochondrial pre-protein transport across the outer mitochondrial membrane. The resulting proteostatic imbalance in the cytosol, if not sufficiently repaired or counteracted by the heat shock response, proteasomal clearance or diminished *de novo* protein synthesis, can induce apoptotic cell death.

known as the mitoprotein-induced stress response and includes several pathways that were previously described independently (Boos et al., 2019 [DOI](#), 2020 [DOI](#)). Here we show that this mitochondrial stress response is highly and specifically induced by a natural lipid compound, t-2-hex, providing evidence and a molecular mechanism of an endogenous modulation of mitochondrial protein import. As a consequence, the accumulated precursor proteins trigger the heat shock response, which increases the production of chaperones and enhances the proteolytic capacity of the ubiquitin-proteasome system (UPS) (Boos et al., 2019 [DOI](#); Weidberg and Amon, 2018 [DOI](#); Wrobel et al., 2015 [DOI](#)). This cellular response to mitochondrial dysfunction is also known as the unfolded protein response activated by mistargeting of proteins (UPRam). The heat shock transcription factor Hsf1 plays a central role in this response, as it is activated not only by heat stress but also during the transition from fermentation to respiration and in the presence of mitochondrial import impairment, all of which result in the accumulation of mitochondrial precursors in the cytosol (Boos et al., 2019 [DOI](#); Groh et al., 2023 [DOI](#); Hahn and Thiele, 2004 [DOI](#)). The sequestration of cytosolic chaperones by these precursor proteins leads to the release and activation of Hsf1. This activation triggers a cascade of gene expressions, including the transcription factor Rpn4, which induces the expression of proteasomal genes to enhance UPS capacity. According to our results, lipidation of the TOM complex at various subunits upon pro-apoptotic conditions blocks the mitochondrial protein entry gate and triggers the heat shock response and proteasomal activity to counteract mislocalized protein accumulation in the cytoplasm.

The external addition of the pro-apoptotic lipid aldehyde t-2-hex is an efficient experimental model to study the induction of apoptosis in various cell types or in purified mitochondria (Amaegberi et al., 2019 [DOI](#); Chipuk et al., 2012 [DOI](#); Kumar et al., 2011 [DOI](#); Manzanares-Estreder et al., 2017 [DOI](#)). The internal trigger of apoptosis comes from the sphingolipid degradation pathway, which is induced upon different cellular stress situations. Genetic manipulations to either favor production of t-2-hex or avoid its further degradation cause mitochondrial damage and inhibit proliferation (Manzanares-Estreder et al., 2017 [DOI](#)). The first enzymatic steps in this pathway from ceramide to sphingosine-1-phosphate and t-2-hex altogether take place in the ER. The t-2-hex detoxifier Hfd1 localizes in foci along the mitochondrial network (Manzanares-Estreder et al., 2017 [DOI](#)) and associates with the outer membrane of the organelle probably by interacting with TOM. Here, Hfd1 would serve as a surveillance factor in order to avoid an impairment of pre-protein import caused by increasing lipidation of the TOM entry gate. This is in line with recent findings, which reveal association of the core TOM complex with an increasing variety of quality control factors such as chaperones, ubiquitin ligases, deubiquitilases and proteases to constantly monitor the protein import process at the mitochondrial surface (Opaliński et al., 2018 [DOI](#); Schulte et al., 2023 [DOI](#)).

Mitochondrial protein import is a tightly monitored traffic hub in eukaryotic cells. Its prolonged malfunction leads to cell death and subtle perturbations cause diseases including neurodegeneration (Coyne et al., 2023 [DOI](#)). Insufficient clearance of stalled mitochondrial precursor proteins from the TOM complex leads to their accumulation in the cytosol and more generally, impairments in mitochondrial protein import lead to the formation of aberrant cytosolic deposits containing high-risk precursor and immature mitochondrial proteins. These deposits further accumulate and cause aggregation of disease-related proteins, including α -synuclein and amyloid β , resulting in cytosolic protein homeostasis imbalance (Nowicka et al., 2021 [DOI](#)). Here we show that this proteostatic imbalance is efficiently created by the t-2-hex lipid aldehyde, which cannot be counteracted by proteasomal clearance given its inefficiency in breaking down protein aggregates. This provides evidence that pro-apoptotic lipid signaling produces a collapse of protein homeostasis via mitochondrial protein import defects. Of note, the capacity of mitochondrial protein import has been recently directly correlated with the clearance of toxic protein aggregates in the cytosol (Ruan et al., 2017 [DOI](#); Schlagowski et al., 2021 [DOI](#)), further strengthening the concept that mitochondrial protein import is a crucial modulator of general proteostasis and specific detoxification of protein aggregates (Maruszczak et al., 2023 [DOI](#)). In line with the results presented

in this study, the levels of pro-apoptotic lipid effectors such as t-2-hex are decisive for the entry into apoptotic death by increasing aggregate formation. Apart from triggering a cell death decision, this balance might also be crucial for cellular lifespan. A response to a mild mitochondrial protein import defect extends lifespan in *Caenorhabditis elegans* (Sladowska et al., 2021 [↗](#)), and more generally, proteasomal activity or a reduced or better controlled protein synthesis rate correlate with longevity in yeast cells (Kruegel et al., 2011 [↗](#); Martinez-Miguel et al., 2021 [↗](#); Mittal et al., 2017 [↗](#); Steffen et al., 2008 [↗](#)). Given that a slowed protein synthesis rate is also beneficial to counteract the pro-apoptotic function of t-2-hex, the action of anti-apoptotic enzymes such as Hfd1 could have effects on cellular lifespan as well. It will be crucial for future studies to evaluate the impact of mitochondrial protein import during lipid-mediated cell death in higher eukaryotic models, which has the potential of significantly broaden our comprehension of the mechanisms leading to intrinsic cell death and their regulation.

Materials and methods

Yeast strains and growth conditions

The yeast strains used in this work are detailed in Supplemental Table 7. Yeast cultures were grown, unless otherwise indicated, at 28°C in Yeast Extract Peptone Dextrose (YPD) media containing 2% glucose, Yeast Extract Peptone Galactose (YPGal) media containing 2% galactose or in Synthetic Dextrose (SD) media containing 0.67% yeast nitrogen base with ammonium sulphate and without amino acids, 50mM succinic acid (pH5.5) and 2% of the respective sugar. According to specific auxotrophies of individual strains, methionine (10mg/l), histidine (10mg/l), leucine (10mg/l) or uracil (25mg/l) were supplemented. Yeast cells were transformed with the lithium acetate/PEG method described by (Gietz and Schiestl, 2007 [↗](#)). Strains expressing particular proteins as Tap-fusions from the chromosomal locus are described in (Ghaemmaghami et al., 2003 [↗](#)). Genomic deletion of particular genes was done with PCR based methods using KAN-MX or his5⁺ (*S. pombe*) containing plasmids pUG6 or pUG27 (Gueldener et al., 2002 [↗](#)). All deletion strains were verified with diagnostic PCR on purified genomic DNA. For genomic promoter swapping, a KAN-MX::TDH3p containing construct was made by cloning the *TDH3* promoter in the pUG6 plasmid at the *EcoRV* and *SpeI* sites. The Hfd1 constitutive overexpressor strain was made by replacing the native promoter with the KAN-MX::TDH3p cassette.

Bioactive lipids t-2-hex and t-2-hex-H₂ were purchased from Sigma and applied directly to yeast cultures from 1mg/ml stock solutions in DMSO. Cycloheximide (Sigma) was added to a final concentration of 250µg/ml from a 50mg/ml stock solution. The ribosomal inhibitor (2-(2-Bromophenyl)-2,3-dihydro-1H-naphtho[1,8-de][1,3,2]diazaborine (DAB) was purchased from TCI Europe and applied to yeast cultures at a final concentration of 20µg/ml from a 20mg/ml stock in ethanol. Mitochondrial uncoupler CCCP was purchased from Sigma and applied directly to yeast cultures to a final concentration of 20µM from stock solutions in DMSO.

Plasmids

The single-copy destabilized reporter plasmid pAG413-GRE2-lucCP⁺ was described previously (Rienzo et al., 2012 [↗](#)). The HFD1p-lucCP⁺ reporter was constructed by cloning the entire *HFD1* promoter (-975/-4) into the *SacI* and *SmaI* sites of pAG413-lucCP⁺ (Rienzo et al., 2012 [↗](#)). For the assay of specific cis-regulatory elements, synthetic oligonucleotides containing three repetitions of PACE and four repetitions of HSE sequences were inserted into the *BspEI* site of plasmid pAG413-CYC1Δ-lucCP⁺ (Rienzo et al., 2012 [↗](#)). The sequences used were the following (consensus sequence for each TF binding site is underlined): PACE, 5'-CCGGCGGTGGCAAAGATATCGGTGGCAAAGTAATCGGTGGCAAAT-3'; HSE, 5'-CCGGCGATATCTTCTAGAAGCTTCTAGAAGT-3'. Other TF-specific real time luciferase reporter plasmids (AP-1, CRE, PDRE) used in this study were described previously (Dolz-Edo et al., 2013 [↗](#);

Vanacloig-Pedros et al., 2019 [DOI](#)). A mitochondrially targeted GFP fusion encoded in the pVT100-mtGFP was used to visualize mitochondrial morphology in yeast cells (Westermann and Neupert, 2000 [DOI](#)).

Quantitative growth assays

For the quantitative estimation of growth parameters, fresh yeast overnight precultures of the indicated yeast strains were diluted in triplicate in the assay medium in multiwell plates to a starting OD₆₀₀=0.1. Growth was then continuously monitored at a wavelength of 600nm every 30min at 28°C on a Tecan Spark multiplate reader for the indicated times and with two rounds of orbital shaking in between reads. The growth curves were processed in Microsoft Excel, and doubling times, lag times and growth yields were calculated with the PRECOG software (Fernandez-Ricaud et al., 2016 [DOI](#)) with respect to control conditions.

Preparation of total protein extracts

Yeast whole cell extracts were generated using the boiling method of alkaline pretreated cells described by (Kushnirov, 2000 [DOI](#)). A total of 2.5 OD₆₀₀ were processed for each individual extract.

Western blot

The following primary antibodies were used for the immunological detection of specific proteins on PVDF membranes: anti-Tap (CAB1001 rabbit polyclonal antibody, Invitrogen, 1:10.000), anti-HA (12CA5 mouse monoclonal antibody, Roche, 1:10.000), anti-Ubiquitin (P4D1 mouse monoclonal antibody, Santa Cruz Biotechnology, 1:1.000), anti-Pgk1 (22C5D8 mouse monoclonal antibody, Invitrogen, 1:5.000). Immunoblots were ECL developed using the secondary antibodies anti-rabbit IgG HRP (Cytiva NA934, 1:10.000) and anti-mouse IgG HRP (Cytiva NA931, 1:10.000). Blots were visualized with an Image Quant LAS4000 system. Total protein loading was visualized with the Direct Blue 71 method described by (Hong et al., 2000 [DOI](#)).

Co-immunoprecipitation experiments

Yeast cells expressing full length Hfd1 fused at the C-term with HA from expression vector pAG426-GPD-ccdB-HA and harboring endogenous genomic fusions of *TOM22* or *TOM70* with the TAP epitope were grown in selective SD medium to OD₆₀₀ = 1. Cells were washed 1x in cold H₂O and 1x in cold buffer A (50mM Tris/HCl pH 7.5, 15mM EDTA, 150mM NaCl, 2mM DTT, 0.1% Triton X-100, 1mM PMSF). Cells were lysed by mechanical disruption with 0.5mm glass beads in buffer A supplemented with complete protease inhibitor cocktail (Roche). Equal amounts of total protein (input samples), quantified by the Bradford method, were incubated with Pan Mouse IgG Dynabeads (Invitrogen) for 2h at 4°C. Beads were extensively washed (5x) with buffer A and finally resuspended in 2x Laemmli buffer. Input and co-precipitated samples were analyzed by anti-HA and anti-TAP immunological detection.

Real-time luciferase assays

Time-elapsd assays with destabilized firefly luciferase reporters in living yeast cells were performed essentially as described before (Rienzo et al., 2012 [DOI](#)). Yeast strains containing the indicated luciferase fusion genes on centromeric plasmids were grown at 28°C in Synthetic Dextrose (SD) medium lacking histidine at pH=3 to low exponential phase. Culture aliquots were then incubated with 0.5mM luciferin (free acid, Synchem, Germany) on a roller at 28°C for 1h. The cells were then transferred in 135µl aliquots to white 96 well plates (Nunc) containing the indicated concentrations of stressor or solvent. The light emission was continuously recorded on a GloMax microplate luminometer (Promega) in three biological replicas. Data were processed with Microsoft Excel. Raw data were normalized for the number of cells in each individual assay and

set to 1 for time point 0. For the comparison of induction folds, luciferase activities are shown throughout this work as relative light units (R.L.U). For the comparison of induction sensitivities, the maximal luciferase activity was set to 100% for the most responsive stressor concentration.

Fluorescence microscopy

Exponentially growing yeast cells on SD media were visualized on a Leica SP8 confocal microscope with a HCXPL APO CS2 63x objective. GFP was visualized with 488nm excitation and 509nm emission wavelengths. For cytoplasmic protein aggregation assays, yeast strains with endogenously GFP-tagged *HSP104* were used. For the visualization of global mitochondrial morphology, yeast strains with plasmid-encoded mtGFP (pVT100-mtGFP) targeted to the mitochondrial lumen were used.

Quantification of proteasome activity

Chymotrypsin-like activity of the proteasome was measured as described previously (Kruegel et al., 2011) with some variations. 10 to 20mL of yeast culture were grown to OD₆₀₀ 1-2 and treated with DMSO, t-2-hex or t-2-hex-H₂ for 3h. Before cell lysis, cells were washed 1x with ice-cold water and 1x with cold lysis buffer (50mM Tris-HCl, pH 7.5, 0.5mM EDTA, 5mM MgCl₂). Cells were resuspended in 500μL of ice-cold lysis buffer supplemented with cOmplete Mini, EDTA-free protease inhibitor cocktail (Roche) and transferred to 2mL tubes with screw cap. After addition of glass beads (0.5mm diameter), cells were lysed in the cold with a Precellys Evolution Touch homogenizer (Bertin Technologies) repeating five cycles of 30s at 10000rpm and 30s in ice. Cell extracts were cleared at 5000g for 5min at 4°C and protein concentration was quantified by the Bradford assay (BioRad). Finally, proteasomal activity was assessed using the Suc-LLVY-AMC substrate (Bachem). 50μg of total protein extract were diluted with lysis buffer in a final volume of 190μL in 96-well black microtiter plates. After addition of 10μL of 2mM stock solution of Suc-LLVY-AMC substrate, fluorescence intensity was measured during 60min at 30°C with a Tecan Spark microplate reader (Tecan) with an excitation wavelength of 380nm and an emission wavelength of 460nm. Chymotrypsin-like activity was calculated as the slope of the linear part of the fluorescence intensity curve and expressed as the % of the Control condition.

RNA seq experiments

30mL of *hfd1Δ* exponentially growing cells were treated in triplicate with DMSO or 100μM t-2-hex for 3h. After treatment, cells were collected, washed with ice-cold water, and frozen in liquid nitrogen. Total RNA extraction was performed using the acid phenol method. Cell pellets were resuspended in 400μL of TES buffer (10mM Tris/HCl pH 7.5, 10mM EDTA, 0.5% SDS) and an equal volume of acid phenol mix (saturated with 0.1M citrate buffer pH 4.3). The mixture was incubated at 65°C for 45min. After 5min on ice, phases were separated by centrifugation (1min, 10000rpm), and the aqueous phase was transferred to a new tube for another extraction with acid phenol. Again, the aqueous phase was transferred to another tube containing 40μL of 3M NaAcetate pH 5.0 and 2.5 volumes of 100% ethanol. Finally, RNA was pelleted by centrifugation at 14000rpm for 10min, washed once with 70% ethanol, and resuspended in 100μL of Milli-Q water. Following extraction, 50μg of total RNA were DNaseI digested and purified using the RNeasy Mini Kit (QIAGEN) according to the manufacturer's instructions. RNA integrity was assessed using a 1% agarose gel.

The samples were processed and massively sequenced by BGI Genomics using the BGISEQ platform (<https://www.bgi.com/us/sequencing-services/rna-sequencing-solutions/transcriptome-sequencing/>). RNA integrity was evaluated using the Agilent Bioanalyzer 2100 (Agilent Technologies) and libraries constructed following DNBSEQ Eukaryotic Stranded Transcriptome library preparation protocol. Samples were sequenced at paired-end 100 bp read length in the DNBSEQ platform. The sequencing data were filtered using SOAPnuke software (version 1.5.2) (Cock et al., 2010) and clean reads were mapped with HISAT2 (version 2.0.4) (Kim et al., 2015).

to the reference genome (*Saccharomyces cerevisiae*_S288C version GCF_000146045.2_R64). On average, about 47.1 million clean reads were generated per sample, and the average mapping ratio with the yeast reference genome was 95.7%, with an average mapping ratio with particular genes of 75.0%. A total of 5851 genes was identified.

All RNAseq samples used in this study are available to download from ArrayExpress (<https://www.ebi.ac.uk/biostudies/arrayexpress/studies>) under dataset accession number E-MTAB-13188. For Gene Ontology (GO) analysis of transcriptomic data, genes upregulated ($>2 \log_2\text{FC}$) and downregulated ($<-2 \log_2\text{FC}$) by t-2-hex were selected. A total of 675 up-regulated genes and 530 down-regulated genes were subjected to GO category enrichment analysis using ShinyGo 0.80 software (<http://bioinformatics.sdstate.edu/go/>) (Ge et al., 2020). The significantly enriched functional groups are summarized in Supplemental Table 1 (up-regulated genes) and Supplemental Table 2 (down-regulated genes).

***In silico* analysis of enriched promoter motifs**

RNAseq raw data were analysed with the Integrated System for Motif Activity Response Analysis (ISMARA) software, powered by the SwissRegulon Portal from the Swiss Institute of Bioinformatics (Balwierz et al., 2014). ISMARA is an online tool for automatically inferring regulatory networks from gene expression data and directly processes raw data of a set of samples and identifies the key transcription factors and miRNAs driving the observed expression state changes.

In this study, RNAseq FASTQ files of the different samples (3x control, 3x t-2-hex) were uploaded to ISMARA web portal (<https://ismara.unibas.ch/mara/>) and analysed against sacSer2 genome assembly. After data processing, results were downloaded for further analysis.

Saturated transposon analysis (SATAY)

SATAY was essentially performed as originally described in (Michel et al., 2017) with some modifications.

Preparation of transposon mutant library: The ByK352 *hfd1Δ::KanMX* strain was transformed with pBK549 plasmid containing the transposon (Michel and Kornmann, 2022). Transformed cells were inoculated into 200mL of SC liquid + 0.2% Glucose + 2% Raffinose -Uracil at OD₆₀₀ of 0.15 and the culture was grown until saturation (OD₆₀₀ = 3.9) for 16 hours. The culture was then concentrated by centrifugation to obtain a final OD₆₀₀ of 39. From this culture, 200μL were plated on 50×8.5cm plates containing 25mL of SC + 2% Galactose -Adenine using glass beads. Plates were incubated for 3 weeks at 30°C. After incubation clones reached a density on the plate of 46750 colonies/plate, on average, with a total number of 2.3E+06 generated mutants. All colonies were then pooled together with very low volume of sterile water. From the pool, two independent cultures of 1L were inoculated in SC +2% Glucose -Adenine. The control one was inoculated at an initial OD₆₀₀ of 0.2 and grown until saturation (OD₆₀₀ of 8) at 30°C for 36 hours. The other culture, used for t-2-hex treatment, was grown for 36 hours from an initial OD₆₀₀ of 0.05 to 3, and then diluted to OD₆₀₀ 0.1 in 500mL of SC +2% Glucose -Adenine treated with 50μM of t-2-hex (Avanti Polar Lipids). After growing for 32 hours to OD₆₀₀ 1.1, it was diluted again to OD₆₀₀ 0.1 in 500mL SC +2% Glucose -Adenine + 50μM of t-2-hex and grown for 24 hours to saturation (OD₆₀₀ 2.9). Finally, cells of both cultures (Control and t-2-hex treatment) were harvested for genomic DNA extraction.

Genomic DNA library preparation: A 500mg cell pellet was resuspended in 500μL of cell breaking buffer (2% Triton X-100, 1% SDS, 100mM NaCl, 100mM Tris-HCl pH 8.0, 1mM EDTA) and divided into 280μL aliquots. To each aliquot, 200μL of Phenol:Chloroform:Isoamyl alcohol (25:24:1) and 300μL of unwashed glass beads (0.5mm) were added. Cells were lysed by subjecting them to 10 cycles of vortexing for 30 seconds followed by incubation on ice for 30 seconds. Next, 200μL of TE buffer was added to each lysate, which was then centrifuged at 16000g for 5 minutes at 4°C. The

upper layer (approximately 400µL) was carefully transferred to a new tube, and 2.5 volumes of 100% ethanol were added, followed by mixing by inversion. The DNA was pelleted by centrifugation at 16000g for 5 minutes at 20°C. The supernatant was discarded, and the DNA pellets were resuspended in 200µL of RNase A solution (250µg/ml) and incubated for 15 minutes at 55°C with gentle agitation using a thermoblock at 1000rpm. Then, 2.5 volumes of 100% ethanol and 0.1 volume of 3M NaOAc pH 5.2 were added. The DNA was pelleted by centrifugation at 16000g for 5 minutes at 20°C. The pellets were washed with 70% ethanol and dried for 10 minutes at 37°C. Finally, the pellets were resuspended in a total volume of 100µL H₂O and incubated for 10 minutes at 55°C with gentle agitation using a thermoblock at 700rpm. The DNA was then analyzed on a 0.6% 1x TBE agarose gel using a standard 1 kb GeneRuler, and quantification was performed using Fiji software.

Library amplification and sequencing: Next, 2µg of both genomic DNA were digested using 50U of *Mbo*I (Thermo Fisher Scientific) and *Hin*1II (Thermo Fisher Scientific) enzymes at 37°C for 24h, in separated reactions. Subsequently, the enzymatic reactions were heat-inactivated at 65°C for 20 minutes. The resulting DNA fragments were circularized by adding 25 Weiss U of T4 ligase (Thermo Fisher Scientific) and ligated at 22°C for 6 hours in a total volume of 400µL. To precipitate the circularized DNA molecules, a mixture of 0.3M sodium acetate (pH 5.2), 1mL of ethanol, and 5µg of linear acrylamide (Ambion AM9520) was applied. The mixture was then incubated at -20°C overnight. After precipitation, the DNA was pelleted at 16000g and 4°C for 20 minutes, followed by washing with 1mL of 70% ethanol. The DNA was then dried at 37°C for 10 minutes. The circularized DNA samples were resuspended in 100µL of water. For transposon amplification, PCR was performed using the forward primer P5_MiniDs (AATGATACGGCGACCAACGAGATCTACTccgtcccgaagttaaataatga) and the reverse primer P7_MiniDs (CAAGCAGAAGACGGCATAACGAGATNNNNNNNNacgaaaacgaacgggataaatac). Each primer consisted of two parts: the Illumina adaptor (uppercase letters) and a fragment from MiniDs (lowercase letters). The 8-nucleotide index present in P7_MiniDs allows distinction among different libraries. For PCR, circularized DNA was divided into 10 x 100µL reactions containing: 10µL of ligated DNA, 10µL dNTPs mix 2mM each, 10µL 10x DreamTaq Buffer (Thermo Fisher Scientific), 1µL 100uM primer P5, 1uL 100µM primer P7, 2µL DreamTaq DNA Polymerase (5U/µL) (Thermo Fisher Scientific). PCR was performed under the following conditions: 95°C 5min, 35x [95°C 30s, 55°C 30s, 72°C 3min], 72°C 10min. Finally, for each condition (Control, t-2-hex) all PCR reactions were pooled into one *Hin*1II-digested pool and one *Mbo*I-digested pool. 100µL from each pool were purified with CleanNGS magnetic beads (CleanNA) according to manufacturer instructions.

Primer Name Sequence (5' → 3')

P5_MiniDs AATGATACGGCGACCAACGAGATCTACTccgtcccgaagttaaataatga

P7_MiniDs_i1 CAAGCAGAAGACGGCATAACGAGAT**CGAGTAAT**acgaaaacgaacgggataaatac

P7_MiniDs_i2 CAAGCAGAAGACGGCATAACGAGATT**TCTCCGGA**acgaaaacgaacgggataaatac

P7_MiniDs_i3 CAAGCAGAAGACGGCATAACGAGATA**AATGAGCG**acgaaaacgaacgggataaatac

P7_MiniDs_i4 CAAGCAGAAGACGGCATAACGAGAT**GGAATCTC**acgaaaacgaacgggataaatac

Purified PCR products were then checked with 4200 TapeStation System (Agilent Technologies) and quantified by fluorimetry with QubitTM dsDNA HS Assay Kit (Invitrogen) and Qubit 4 Fluorometer (Invitrogen). Equal amounts of *Mbo*I-digested and *Hin*1II-digested libraries from both Control and t-2-hex samples, were pooled, diluted to 20pM, and sequenced at single 75bp read length using MiSeq Reagent Kit v3 150-cycles (Illumina) on the MiSeq platform (Illumina), according to manufacturer instructions. 3.4µL of 100µM primer 688_minidsSEQ1210

(TTTACCGACCGTTACCGACCGTTTTTCATCCCTA) and 600µL of 0.5µM primer Custom_index1 (GGTTTTTCGATTACCGTATTATCCCGTTCGTTTTTCGT) were loaded into wells 12 and 19 of the sequencing cartridge, respectively.

Sequences alignment and determination of transposon insertion sites and corresponding read numbers: The FastQ files were analysed, aligned to the *sacCer3* genome and processed using R and Bioconductor packages ShortRead, Rsamtools and Rbowtie ((Michel and Kornmann, 2022 [DOI](#)), Supplement File 1 and Supplement File 2). Volcano plots were generated from two technical replicates of the sequencing. All SATAY data from this study are deposited to download from ArrayExpress (<https://www.ebi.ac.uk/biostudies/arrayexpress/studies> [DOI](#)) under dataset accession number E-MTAB-13208.

t-2-hex-alkyne *in vitro* lipidation assays

Lipidation assays with the t-2-hex-alkyne probe were essentially performed as described previously (Jarugumilli et al., 2018 [DOI](#)). Yeast cells expressing HA-tagged Tom40 from the natural promoter on centromeric plasmids (Becker et al., 2011 [DOI](#)) or control cells were grown to exponential growth phase in Yeast Extract Peptone Glycerol/Ethanol (YPGE) medium containing 3% glycerol/2% ethanol. Cells were washed 1x with cold H₂O and 1x with cold TSB buffer (10mM Tris/HCl pH 7.5, 0.6M sorbitol). Cells were lysed by mechanical rupture with 0.5mm glass beads in TSB buffer supplemented with 1mM PMSF and complete protease inhibitor cocktail (Roche). The mitochondrial fraction was enriched from the lysates by two rounds of differential centrifugation: 5min at 3500g discarding the pellet and subsequent centrifugation of the supernatant at 16000g for 10min. The final mitochondria enriched pellet was resuspended in 250µl of Hepes lysis buffer (50mM Hepes pH 7.4, 0.5% NP-40) supplemented with complete protease inhibitor cocktail (Roche). Total protein was quantified with the Bradford method, and the mitochondrial fractions adjusted to 0.1mg/ml in Hepes lysis buffer. 500µl aliquots were used for *in vitro* lipidation experiments. The indicated concentrations of t-2-hex-alkyne ((E)-2-Hexadecenyl Alkyne, 20714 Cayman Chemicals) or solvent alone were added from a 1mg/ml stock in ethanol and the samples were rotated at room temperature for 1h. In the thiol competition experiments, 1mM DTT was added to the samples before addition of t-2-hex-alkyne. After removal of the input samples (100µl), the rest of the samples was treated with the click reaction mix (40µl) of 100µM TBTA (678937 Sigma), 1mM TCEP (C4706 Sigma), 1mM CuSO₄, 100µM Azide-PEG3-biotin conjugate (762024 Sigma) for 1h at room temperature. Samples were finally precipitated by adding 3.4ml chilled methanol over night at -80°C. Proteins were pelleted by centrifugation at 17000g for 15min and dried in a Speed Vac centrifuge for 5min. Proteins were first dissolved by addition of 50µl PBST (8mM Na₂HPO₄, 150mM NaCl, 2mM KH₂PO₄, 3mM KCl, 0.05% Tween 20, pH 7.4) containing 1% SDS and then diluted in 950µl PBST. Protein mixtures were incubated with 40µl PBST washed Streptavidin agarose resin (50% slurry, 69203-3 Merck) and incubated for 2h at room temperature on a roller. Agarose beads were washed 3x with PBST and proteins eluted with 80µl of elution buffer (95% formamide, 10mM EDTA) for 5min at 95°C. The supernatant was supplied with 1x Laemmli buffer and proteins were analyzed by immunoblotting with anti-HA antibodies.

For the chemoproteomic analysis of t-2-hex targets in intact yeast mitochondria, the mitochondria enriched fraction was prepared from wild type cells as described above. A total of 50µg of mitochondrial fraction was treated with 100µM or 10µM t-2-hex-alkyne or solvent alone and the samples were rotated at room temperature for 1h. Samples were click chemistry treated and methanol precipitated as described above. Dried protein pellets were dissolved first in 500µl in resuspension buffer containing 50mM Tris/HCl pH7.4, 150mM NaCl, 10mM EDTA, 1% SDS. Samples were further diluted by adding 500µl of resuspension buffer containing 50mM Tris/HCl pH7.4, 150mM NaCl, 10mM EDTA, 1% Brij97. 40µl of PBS washed Streptavidin agarose beads were added to the samples and incubation was performed on a rotator at room temperature for 1.5h. Beads were washed 3x with PBS + 0.2% SDS and 1x with 250mM ammonium bicarbonate. After removing the supernatant, beads were incubated with 500µl of 8M urea, 50µl of 500mM TCEP and 50µl of 400mM iodoacetamide and incubated on a rotator in the dark for 40min. Finally beads were

washed 2x with 250mM ammonium bicarbonate. Further sample preparation and proteomic analysis was performed in the proteomics facility of SCSIE University of Valencia, a member of Proteored. Proteins were digested on the beads with trypsin over night. Tandem mass spectrometry analysis (LC-MS/MS) was performed in a Tims TOF fleX mass spectrometer (Bruker). The PASER system (Bruker) was used to search the MS and MSMS data with the Sequest algorithm (ProLuCID) with the following parameters: SwissProt 23.03.10 database; Trypsin specificity; IAM cys-alkylation and taxonomy not restricted.

An excel document was generated with all the proteins identified with $FDR \leq 1\%$. Proteins were ranked according to their number of spectra assigned to the protein in the t-2-hex-alkyne versus mock treated samples. Differential enrichment analysis of the proteomic data was performed with the amica software (Didusch et al., 2022 [DOI](#)). Proteins were ranked according to their $\log_2$ fold induction comparing lipid- and mock-treated samples with a threshold of ≥ 1.5 , and the adjusted p-value was calculated. A total of 217 (100 μ M t-2-hex) and 459 (10 μ M t-2-hex) differentially enriched proteins were identified.

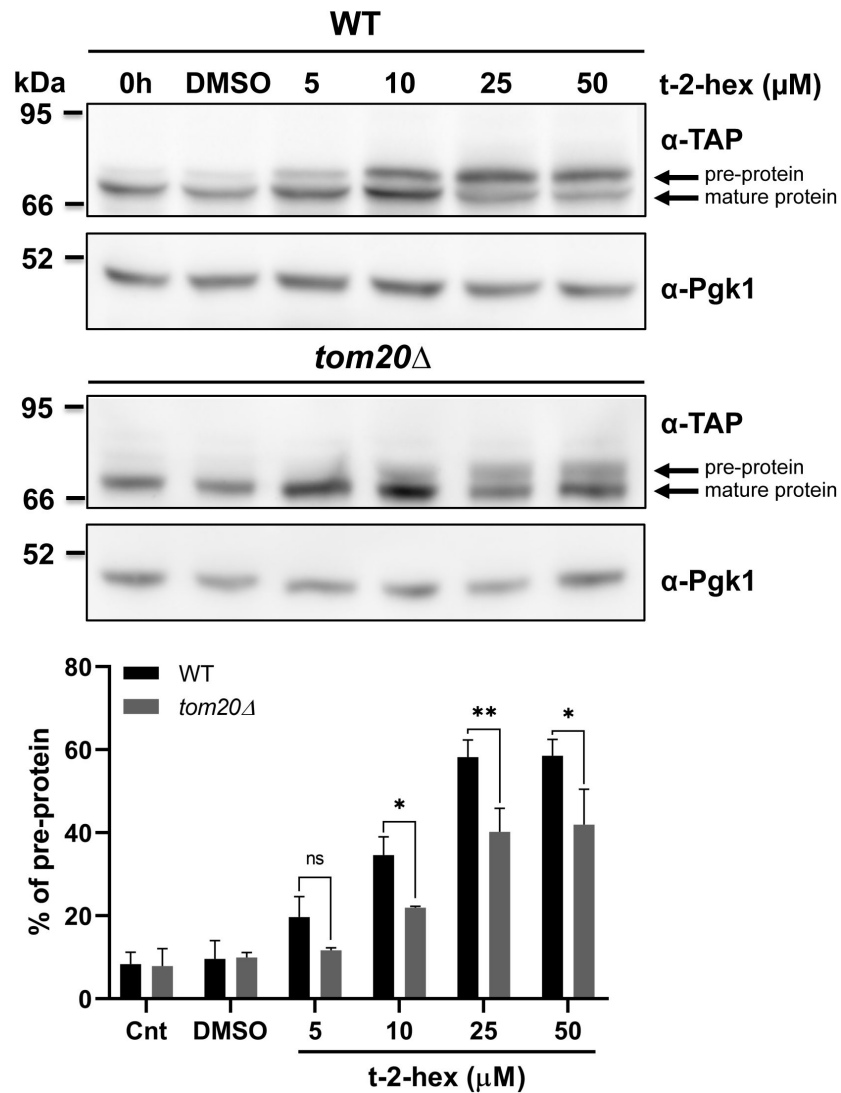
Statistical analyses

Data are presented as mean $\pm$ standard error mean. The number of biological replicas (n) is given for each experiment in the figure legends. Comparisons between different groups were performed by the Student's t test. All analyses were performed with GraphPad Prism 8.0.1 software (GraphPad Software, Inc.). A p-value < 0.05 was considered statistically significant.

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Supplementary



Supplemental Figure 1.

Tom20 function determines the extent of t-2-hex induced pre-protein accumulation. Upper panels: Aim17 pre-protein accumulation in response to different t-2-hex concentrations was quantified in wild type and *tom20Δ* cells. Cells were either untreated, or treated with the indicated lipid concentrations or DMSO for 20 min. The percentage of pre-protein relative to total Aim17 protein was calculated in the lower panel ($n = 3$). * $p < 0.05$, ** $p < 0.01$ by Student's unpaired t-test, ns = not significant.

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Editors

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Reviewer #2 (Public Review):

This study elucidates the toxic effects of the lipid aldehyde trans-2-hexadecenal (t-2-hex). The authors show convincingly that t-2-hex induces a strong transcriptional response, leads to proteotoxic stress and causes the accumulation of mitochondrial precursor proteins in the cytosol.

The data shown are of high quality and well-controlled. The genetic screen for mutants that are hyper-and hypo-sensitive to t-2-hex is elegant and interesting, even if the mechanistic insights from the screen are rather limited. Moreover, the authors show evidence that t-2-hex affects subunits of the TOM complex. However, they do not formally demonstrate that the lipidation of a TOM subunit is responsible for the toxic effect of t-2-hex. A t-2-hex-resistant TOM mutant was not identified. Nevertheless, this is an interesting and inspiring study of

high quality. The connection of proteostasis, mitochondrial biogenesis and sphingolipid metabolism is exciting and will certainly lead to many follow-up studies.

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Reviewer #3 (Public Review):

Summary: The authors investigate the effect of high concentrations of the lipid aldehyde trans-2-hexadecenal (t-2-hex) in a yeast deletion strain lacking the detoxification enzyme. Transcriptomic analyses as global read out reveal that a large range of cellular functions across all compartments are affected (transcriptomic changes affect 1/3 of all genes). The authors provide additional analyses, from which they built a model that mitochondrial protein import caused by modification of Tom40 is blocked.

Strengths:

Global analyses (transcriptomic and functional genomics approach) to obtain an overview of changes upon yeast treatment with high doses of t-2-hex.

Weaknesses:

The use of high concentrations of t-2-hex in combination with a deletion of the detoxifying enzyme Hfd1 limits the possibility to identify physiological relevant changes. From the hundreds of identified targets the authors focus on mitochondrial proteins, which are not clearly comprehensible from the data. The main claim of the manuscript that t-2-hex targets the TOM complex and inhibits mitochondrial protein import is not supported by experimental data as import was not experimentally investigated. The observed accumulation of precursor proteins could have many other reasons (e.g. dissipation of membrane potential, defects in mitochondrial presequence proteases, defects in cytosolic chaperones, modification of mitochondrial precursors by t-2-hex rendering them aggregation prone and thus non-import competent). However, none of these alternative explanations have been experimentally addressed or discussed in the manuscript. Furthermore, many of the results have been reported before (interaction of Tom22 and Tom70 with Hfd1) or observed before (TOM40 as target of t-2-hex in human cells).

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Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public Review):

Summary:

Fita-Torró et al. study the toxic effects of the intermediary lipid degradation product trans-2-hexadecenal (t-2-hex) on yeast mitochondria and suggest a mechanism by which Hfd1 safeguards Tom40 from lipidation by t-2-hex and its consequences, such as mitochondrial protein import inhibition, cellular proteostasis deregulation, and stress-responses.

The authors aimed to dissect a mechanism for t-2-hex' apoptotic consequences in yeast and they suggest it is via lipidation of Tom40 but really under the tested conditions everything seems lipidated. Thus, it is unclear whether Tom40 is the crucial causal target. They also do not provide much biochemical experiments to investigate this phenomenon

further functionally. Tom40 is one possible and perhaps, given the cellular consequences, a reasonable candidate but not validated beyond in vitro lipidation by exogenous t-2-hex.

In the revised version of our manuscript, we have now included extensive new experimentation, which shows that protein import at the TOM complex is a physiologically important target of the pro-apoptotic lipid t-2-hex and that enzymes such as the Hfd1 dehydrogenase sensitively regulate this inhibition. In vitro chemoproteomic experiments have now been performed at more physiological t-2hex concentrations of 10 μ M, which is lower than published data in human cell models. Consistently, several TOM and TIM subunits are enriched in these in vitro lipidation studies (new Fig. 8B). Tom40 lipidation alone is not sufficient to explain t2-hex toxicity, as a cysteine-free version of Tom40 does not confer tolerance to the apoptotic lipid (new Fig. 8D). Importantly however, the loss of function of nonessential accessory Tom subunits 70 or 20 confers t-2-hex tolerance (new Fig. 8D) indicating that pre-protein import at the TOM complex is a physiological target of t2-hex most likely dependent on lipidation of more Tom subunits than just the essential Tom40 pore. Moreover, we now show that mitochondrial protein import is inhibited by the lipid at low physiological doses of 10 μ M and that this inhibition is modulated by the gene dose of the t-2-hex degrading Hfd1 enzyme (new Fig. 5G).

Strengths:

The effects of lipids and their metabolic intermediates on protein function are understudied thus the authors' research contributing to elucidating direct effects of a single lipid is appreciated. It is particularly unknown by which mechanism t-2hex causes cell death in yeast. The authors elegantly use modulation of the levels of enzyme Hfd1 that endogenously catabolizes t-2-hex as an approach to studying t2-hex stress. Understanding the cause and consequences of this stress is relevant for understanding fundamental regulation mechanisms, and also to human health since the human homolog of Hfd1, ALDH3A2, is mutated in Sjögren-Larsson Syndrome. The application of a variety of global transcriptomic, functional genomic, and chemoproteomic approaches to study t-2-hex stress targets in the yeast model is laudable.

Weaknesses:

- The extent of the contribution of Tom40 lipidation to the general t-2-hex stress phenotype is unclear. Is Tom40 lipidation alone enough to cause the phenotype? An alteration of the cysteine residue in question could help answer this key question.

Deletion of all four cysteine residues in Tom40 is not sufficient to confer resistance to t-2-hex stress. This result had been included in the original manuscript, but was somehow hidden in the Discussion. The revised manuscript now includes t-2hex tolerance assays for the Tom40 cysteine free mutant in new Figure 8. As a result, cysteine lipidation of Tom40 alone is not sufficient to confer t-2-hex toxicity. This implies most likely other lipidation targets within the TOM and TIM complexes, as indicated by our in vitro lipidation studies. We therefore included the non-essential adaptor proteins Tom70 and Tom20 of the TOM complex and tested the tolerance of the respective deletion mutants in t-2-hex tolerance assays. As shown in new Figure 8, the absence of Tom70 and Tom20 function significantly increases tolerance to t-2hex and the tom20 mutant accumulates less Aim17 pre-protein upon t-2-he stress, indicating that the TOM complex is a physiologically important target of the proapoptotic lipid, which acts most likely via lipidation of more subunits than the Tom40 import channel.

- It is unclear whether the exogenously applied amounts of t-2-hex (concentrations chosen between 25-200 μ M) are physiologically relevant in yeast cells. For comparison, Chipuk et al. (2012) used at most 1 μ M on mitochondria of human cells, while Jarugumilli

et al. (2018) considered 25 μ M a 'lower dose' on human cells. Since the authors saw responses below 10 μ M (Fig. 3B) and at the lowest selected concentration of 25 μ M (Fig. 8), why were no lower, likely more specific, concentrations applied for the global transcriptomic and chemoproteomic experiments? Key experiments have to be repeated with the lower concentrations.

We have now performed several experiments with lower t-2-hex concentrations. A new chemoproteomic study with 10 μ M t-2-hex-alkyne has been conducted and the new results added to the supplementary information, combining 10 μ M and 100 μ M in vitro lipidation studies (Suppl. Table 6). Many subunits of the TOM and TIM complexes consistently are enriched significantly in both chemoproteomic experiments. These new data are summarized in revised Figure 8. Additionally we have performed in vivo pre-protein assays with lower t-2-hex concentrations. As shown in new Figure 5, Aim17 mitochondrial import is already inhibited by t-2-hex doses as low as 10 μ M in a wild type strain, and that this inhibition is enhanced in a hfd1 mutant and alleviated in a Hfd1 overexpressor. It is important to note that a dose of 10 μ M of external t-2-hex addition is significantly lower than doses applied to human cell cultures such as in Jarugumilli et al. (2018). It proves that mitochondrial protein import is a sensitive and physiologically relevant t2-hex target in our yeast models and that t-2-hex detoxification by enzymes such as the Hfd1 dehydrogenase sensitively regulates this specific inhibition.

- The amount of t-2-hex applied is especially important to consider in light of over 1300 proteins lipidated to an extent equal to or greater than Tom40 (Supp. Table 6). This chemoproteomic experiment (Fig. 8B, Supp. Table 6) is also weakened by the inclusion of only 2 replicates, thus precluding assessment of statistical significance. The selection of targets in Fig. 8B as "among the best hits" is neither immediately comprehensible nor further explained and represents at best cherry-picking. Further evidence based on statistical significance or validation by other means should be provided.

We performed the chemoproteomic screens as described by Jarugumilli et al. (2018) with 2 replicates of mock treated versus 2 replicates of t-2-hex-alkyne treated cell extracts. A new chemoproteomic study with 10 μ M t-2-hex-alkyne has been conducted and the new results added to the supplementary information combining 10 μ M and 100 μ M in vitro lipidation studies (Suppl. Table 6). Differential enrichment analysis of the proteomic data was performed with the amica software (Didusch et al., 2022). Proteins were ranked according to their log2 fold induction comparing lipid- and mock-treated samples with a threshold of ≥ 1.5 , and the adjusted p-value was calculated. Several TOM and TIM subunits were consistently identified as differentially enriched proteins, which is summarized in new Figure 8B.

- The authors unfortunately also underuse the possible contribution of mass spectrometry technology to in addition determine the extent and localization of lipidation on a global scale (especially relevant since Cohen et al. (2020) suggest site-specific mechanisms).

We agree that site-specific modifications of t-2-hex will be most likely important in the inhibition or other type of regulation of specific target proteins. Our collective data show that in the case of the inhibition of mitochondrial protein import, several lipidation events on TOM and TIM are involved. Dissection of individual cysteine lipidations on those subunits will be interesting, but we feel that this is out of the scope of the present work.

- The general novelty of studying t-2-hex stress is lowered in light of existing literature in humans (see e. g. Chipuk et al., 2012; Cohen et al., 2020; Jarugumilli et al., 2018), and in yeast by the same authors (Manzanares-Estreded et al., 2017) and as the authors

comment themselves, a significant part of the manuscript may represent rather a confirmation of the already described consequences of t-2-hex stress

We do not agree and we have not commented that our present study is a mere confirmation of t-2-hex stress previously applied in yeast and human models. In humans, t-2-hex has been identified as an efficient pro-apoptotic lipid, which causes mitochondrial dysfunction via direct lipidation of Bax, however the studies of Jarugumilli et al. (2018) revealed that many other direct t-2-hex targets exist, which remained uninvestigated to date. This work continues our previous studies (Manzanares-Estreded et al., 2017), where we show that t-2-hex is a universal proapoptotic lipid applicable in yeast models and contributes important novel findings, such as the massive transcriptional response resembling proteostatic defects caused by t-2-hex, mitochondrial protein import as a physiologically important and direct target of t-2-hex, the function of detoxifying enzymes such as Hfd1 in modulating lipid-mediated inhibition of mitochondrial protein import and general proteostasis. Additionally, we provide transcriptomic, chemoproteomic and functional genomic data to the scientific community, which will be a rich source for future studies on yet undiscovered pro-apoptotic mechanisms employed by t-2-hex.

Reviewer #2 (Public Review):

This study elucidates the toxic effects of the lipid aldehyde trans-2-hexadecenal (t-2-hex). The authors show convincingly that t-2-hex induces a strong transcriptional response, leads to proteotoxic stress, and causes the accumulation of mitochondrial precursor proteins in the cytosol.

The data shown are of high quality and well controlled. The genetic screen for mutants that are hyper- and hypo-sensitive to t-2-hex is elegant and interesting, even if the mechanistic insights from the screen are rather limited. The last part of the study is less convincing. The authors show evidence that t-2-hex affects subunits of the TOM complex. However, they do not formally demonstrate that the lipidation of a TOM subunit is responsible for the toxic effect of t-2-hex. A t-2-hex-resistant TOM mutant was not identified. Moreover, it is not clear whether the concentrations of t-2-hex in this study are physiological. This is, however, a critical aspect. The literature is full of studies claiming the toxic effects of compounds such as H₂O₂; even if such studies are technically sound, they are misleading if nonphysiological concentrations of such compounds were used.

Nevertheless, this is an interesting study of high quality. A few specific aspects should be addressed.

We have now performed t-2-hex toxicity assays using several mutants in Tom subunits, the cysteine free mutant of the essential Tom40 core channel and deletion mutants in the accessory subunits Tom70 and Tom20 (new Figure 8). As a result, cysteine lipidation of Tom40 alone is not sufficient to confer t-2-hex toxicity. This implies most likely other lipidation targets within the TOM and TIM complexes, as indicated by our in vitro lipidation studies. Indeed, as shown in new Figure 8, the absence of Tom70 and Tom20 function significantly increases tolerance to t-2-hex indicating that the TOM complex is a physiologically important target of the proapoptotic lipid, which acts most likely via lipidation of more subunits than the Tom40 import channel.

We have now performed several experiments with lower t-2-hex concentrations. A new chemoproteomic study with 10 μ M t-2-hex-alkyne has been conducted and the new results added to the supplementary information combining 10 μ M and 100 μ M in vitro lipidation studies (Suppl. Table 6). Many subunits of the TOM and TIM complexes consistently are enriched significantly in both chemoproteomic experiments. These new data are summarized in revised Figure 8.

Additionally we have performed in vivo pre-protein assays with lower t-2-hex concentrations. As shown in new Figure 5, Aim17 mitochondrial import is already inhibited by t-2-hex doses as low as 10 μ M in a wild type strain, and that this inhibition is enhanced in a hfd1 mutant and alleviated in a Hfd1 overexpressor. It is important to note that a dose of 10 μ M of external t-2-hex addition is significantly lower than doses applied to human cell cultures such as in Jarugumilli et al. (2018). It proves that mitochondrial protein import is a sensitive and physiologically relevant t2-hex target in our yeast models and that t-2-hex detoxification by enzymes such as the Hfd1 dehydrogenase sensitively regulates this specific inhibition.

Reviewer #3 (Public Review):

Summary: The authors investigate the effect of the lipid aldehyde trans-2hexadecenal (t-2-hex) in yeast using multiple omic analyses that show that a large range of cellular functions across all compartments are affected, e.g. transcriptomic changes affect 1/3 of all genes. The authors provide additional analyses, from which they built a model that mitochondrial protein import caused by modification of Tom40 is blocked.

Strengths: Global analyses (transcriptomic and functional genomics approach) to obtain an unbiased overview of changes upon t-2-hex treatment.

Weaknesses: It is not clear why the authors decided to focus on mitochondria, as only 30 genes assigned to the GO term "mitochondria" are increasing, and also the follow-up analyses using SATAY is not showing a predominance for mitochondrial proteins (only 4 genes are identified as hits). The provided additional experimental data do not support the main claims as neither protein import is investigated nor is there experimental evidence that lipidation of Tom40 occurs in vivo and impacts on protein translocation.

30 mitochondrial gene functions are very strongly (>10 fold) up-regulated by t-2-hex. However, when genes up-regulated (>2 log2FC) or down-regulated (<-2 log2FC) by t-2-hex were selected and subjected to GO category enrichment analysis, we found that "Mitochondrial organization" was the most numerous GO group activated by t-2-hex, while it was "Ribosomal subunit biogenesis" for t-2-hex repression (new data in Suppl. Tables 1 and 2).

In the revised version of our manuscript, we have now included extensive new experimentation, which shows that protein import at the TOM complex is a physiologically important target of the pro-apoptotic lipid t-2-hex and that enzymes such as the Hfd1 dehydrogenase sensitively regulate this inhibition. In vitro chemoproteomic experiments have now been performed at more physiological t-2hex concentrations of 10 μ M, which is lower than published data in human cell models. Consistently, several TOM and TIM subunits are enriched in these in vitro lipidation studies (new Fig. 8B). Tom40 lipidation alone is not sufficient to explain t2-hex toxicity, as a cysteine-free version of Tom40 does not confer tolerance to the apoptotic lipid (new Fig. 8D). Importantly however, the loss of function of nonessential accessory Tom subunits 70 or 20 confers t-2-hex tolerance (new Fig. 8D) indicating that pre-protein import at the TOM complex is a physiological target of t2-hex most likely dependent on lipidation of more Tom subunits than just the essential Tom40 pore. Moreover, we now show that mitochondrial protein import is inhibited by the lipid at low physiological doses of 10 μ M and that this inhibition is modulated by the gene dose of the t-2-hex degrading Hfd1 enzyme (new Fig. 5G).

Recommendations for the authors:

Reviewer #1 (Recommendations For The Authors):

Private recommendations for the authors

- On the existing data from Supp. Table 6, the authors may include a global assessment of whether or not the protein included a cysteine (the likely site for lipidation).

Although free cysteines in target proteins are the most frequent sites of modification by LDEs such as t-2-hex, other amino acids such as lysines or histidines can be lipidated by these lipid derivatives. Therefore we would like to exclude this information from our chemoproteomic data.

- What determines whether a gene is labeled in Fig. 6B other than fold change? Why is *MAC1* with the highest FC not shown?

We analyzed the potential anti-apoptotic SATAY hits with a $\log_2 < -0.75$ according to expected detoxification pathways (heat shock response, pleiotropic drug response), to their function in the ER (the intracellular site where t-2-hex is generated) or in mitochondria (the major t-2-hex target identified so far). This is now better described in the text. As for the potential pro-apoptotic SATAY hits, we analyzed gene functions with a $\log_2 > 1.5$ and marked the predominant groups “Cytosolic ribosome and translation” and “Amino acid metabolism”. In any case, the interested reader has all SATAY data available in supplemental tables 4 and 5 to find alternative gene functions with a potential role in cellular adaptation to t-2-hex.

- Supplementary Table numbering should be double-checked.

Ok, numbering has been double-checked.

Reviewer #2 (Recommendations For The Authors):

Major points

(1) Identification of the t-2-hex target. Neither Tom70, Tom20 nor the cysteine in Tom40 is essential. If one of these components is critical for the t-2-hex-mediated toxicity, mutants should be t-2-hex-resistant. This is a straight-forward, simple, and critical experiment.

We have now performed t-2-hex toxicity assays in the cysteine free Tom40 mutant, and tom20 and tom70 deletion mutants. As shown in new Figure 8, cysteine lipidation of Tom40 alone is not sufficient to confer t-2-hex toxicity. However, the absence of Tom70 and Tom20 function significantly increases tolerance to t-2-hex indicating that the TOM complex is a physiologically important target of the proapoptotic lipid, which acts most likely via lipidation of more subunits than the Tom40 import channel.

(2) The authors claim that t-2-hex blocks the TOM complex. Since *in vitro* import assays with yeast mitochondria are a well established and simple technique, the authors should isolate mitochondria from their cells and perform import experiments. It is expected that those mitochondria show reduced import rates, however, swelling of these mitochondria to mitoplasts should suppress the import defect.

We agree that our study does not investigate a direct effect of t-2-hex on the import capacity of purified mitochondria. However, we determine the *in vivo* accumulation of several mitochondrial precursor proteins, which is widely used to assay for the efficiency of mitochondrial protein import, for example the recent hallmark paper discovering the mitoCPR protein import surveillance pathway exclusively uses epitope-tagged mitochondrial precursors to determine the regulation of mitochondrial protein import (Weidberg and Amon, Science 2018 360(6385)). Additionally, our new results that mutants in accessory TOM subunits 20 and 70 are hyperresistant to t-2-hex (Figure 8D) and that deletion of TOM20

decreases the t-2-hex induced pre-protein accumulation (Suppl. Figure 1) identify the TOM complex and hence protein import at the outer mitochondrial membrane as a physiologically important t-2-hex target.

(3) The first part of the study is very strong. The last figure is also of good quality, however, it is not clear whether the effects on TOM subunits are really causal for the observed t-2-hex effect on gene expression. The authors might cure this by improved data or by avoiding bold statements such as: 'Hfd1 associates with the Tom70 subunit of the TOM complex and t-2-hex covalently lipidates the central Tom40 channel, which altogether indicates that transport of mitochondrial precursor proteins through the outer mitochondrial membrane is directly inhibited by the pro-apoptotic lipid and thus represents a hotspot for pro- and anti-apoptotic signaling.' (Abstract).

We now show that several TOM and TIM subunits are lipidated in vitro by physiological low t-2-hex concentrations, that loss of function of accessory subunits Tom20 or Tom70 rescues t-2-hex toxicity (new Figure 8) and that the gene dose of Hfd1 determines the degree of mitoprotein import block (new Figure 5). These data identify the TOM complex as a physiologically important target of the pro-apoptotic lipid. The Abstract has been modified accordingly.

(4) If the t-2-hex levels are in a physiological range, one would expect that overexpression of Hfd1 prevents the t-2-hex-induced import arrest.

We have now confirmed that overexpression of Hfd1 indeed prevents inhibition of mitochondrial protein import by t-2-hex. As shown in new Figure 5, Aim17 mitochondrial import is already inhibited by t-2-hex doses as low as 10 μ M in a wild type strain, and that this inhibition is enhanced in a hfd1 mutant and alleviated in a Hfd1 overexpressor.

(5) The authors claim that Fmp52 is a t-2-hex-detoxifying enzyme, but do not show evidence. They should rewrite this sentence and be more cautious, or they should show that increased Fmp52 levels indeed deplete t-2-hex from mitochondria.

We show that loss of Fmp52 function leads to a strong t-2-hex sensitivity. Fmp52 belongs to the NAD-binding short-chain dehydrogenase/reductase (SDR) family and localizes to highly purified mitochondrial outer membranes (Zahedi et al, 2006). These are the indications that suggest that Fmp52 participates in the enzymatic detoxification of t-2-hex in addition to Hfd1. The Results section has been modified accordingly.

Minor points:

(6) Aim17 was recently identified as a characteristic constituent of cytosolic protein aggregates named MitoStores (Krämer et al., 2023, EMBO J). The authors might test whether the cytosolic Aim17 protein colocalizes with the Hsp104-GFP granules that accumulate upon t-2-hex exposure as shown in Fig. 4A.

We agree that determining the fate of unimported mitochondrial precursors upon t-2-hex stress would be interesting. We have made some attempts to co-visualize Aim17-dsRed and Hsp104-GFP upon t-2-hex treatment, but we still have some technical issues. While we clearly see that Aim17 accumulates in cytoplasmic foci upon prolonged t-2-hex exposure, we are not able to determine colocalization with Hsp104, in great part because t-2-hex causes mitochondrial fragmentation, which leads to the appearance of Aim17-stained foci in the cytosol independently of protein aggregates. While so far we are not able to localize Aim17 unambiguously in Hsp104 containing aggregates (mitoStores) upon lipid stress, we would like to move the manuscript farther without those experiments.

(7) In Fig. 1A, the figures of the different lines are difficult to distinguish. Lines of one color with different intensities would be better suited.

We have been working before with dose-response profiles generated by the destabilized luciferase system and found that the color-coded representation of the plots is the most effective way to represent the data, see for example Fita-Torró et al. Mol Ecol. 2023 32(13):3557-3574, Pascual-Ahuir et al. BBA 2019 1862(4):457-471, Rienzo et al., Mol Cell Biol. 2015 35(21):3669-83, and several other publications. Therefore we want to keep the format of the Figure.

(8) A title page should be added to each of the supplemental data files with short descriptions of the information that is provided in the columns of the tables. Response: Explanatory title pages have been now added to the supplemental data files.

Reviewer #3 (Recommendations For The Authors):

Figure 5A: The authors aim to assess protein import, however, their experimental set-up is not suited and does not allow conclusions about protein translocation into mitochondria. The authors monitor protein steady state levels, which does not reflect import capacity. For this e.g. pulse-chase experiments coupled to coIP or in organello import assays with radiolabeled substrate proteins would be required. In addition, the authors lack a non-treated control to show that no precursor accumulates in the absence of CCCP and t-2-hex. At the moment, the conclusion of blocked import cannot be made, as there are many other explanations for the observed steady state levels, e.g. the TAP tag interfered with the import competence of the precursor or t-2-hex could impact on MPP function (in particular as Figure 8B shows that also intra-mitochondrial proteins undergo modification by t-2-hex).

We agree that our study does not investigate a direct effect of t-2-hex on the import capacity of purified mitochondria. However, we determine the in vivo accumulation of several mitochondrial precursor proteins, which is widely used to assay for the efficiency of mitochondrial protein import, for example the recent hallmark paper discovering the mitoCPR protein import surveillance pathway exclusively uses epitope-tagged mitochondrial precursors to determine the regulation of mitochondrial protein import (Weidberg and Amon, Science 2018 360(6385)). Figure 5 contains several non-treated control experiments, which show that no (or less in the case of Ilv6) precursors of Tap-tagged Aim17, Cox5a, Ilv6, or Sdh4 accumulate in the absence of CCCP or t-2-hex. This is shown in Figure 5A for untreated cells or in Figure 5B and new Figure 5G for solvent (DMSO) treated cells. This demonstrates that the Tap-tag does not interfere with the import of the respective precursors. Additionally, our new results that mutants in accessory TOM subunits 20 and 70 are hyperresistant to t-2-hex (Figure 8D) identify the TOM complex and hence protein import at the outer mitochondrial membrane as a physiologically important t-2-hex target.

Figure 8: The conclusion that Tom40 is directly lipidated comes from an in vitro assay, with the conclusion that Tom40 is the main target, because it is the only Tom protein with a cysteine (Tom70 as not being part of the Tom core is excluded, however, lack of Tom70 function would also have detrimental consequences for mitochondrial protein import). However, there is no experiment showing a modification of Tom40 and a consequence for protein import. The proposed model is therefore very far-fetched and several aspects are speculation but not supported by experimental data. To propose such a model, the author needs to show experimental evidence, e.g. by generating a yeast strain in which the cysteine i Tom40 are replaced by e.g. Serine residues, and then assess if protein import (e.g. pulse-chase assays) are not affected anymore upon addition of t-2-hex.

Deletion of all four cysteine residues in Tom40 is not sufficient to confer resistance to t-2-hex stress. This result had been included in the original manuscript, but was somehow hidden in the Discussion. The revised manuscript now includes t-2-hex tolerance assays for the Tom40 cysteine free mutant in new Figure 8D. As a result, cysteine lipidation of Tom40 alone is not sufficient to confer t-2-hex toxicity. This implies most likely other lipidation targets within the TOM and TIM complexes, as indicated by our in vitro lipidation studies. We therefore included the non-essential adaptor proteins Tom70 and Tom20 of the TOM complex and tested the tolerance of the respective deletion mutants in t-2-hex tolerance assays. As shown in new Figure 8D, the absence of Tom70 and Tom20 function significantly increases tolerance to t-2-hex indicating that the TOM complex is a physiologically important target of the pro-apoptotic lipid, which acts most likely via lipidation of more subunits than the Tom40 import channel.

Figure 8A: The pulldown experiments lack positive (other Tom subunits) and negative controls and were performed with (large) tags on all proteins, which can easily result in false positive interactions. The conclusion that Hfd1 interacts with Tom70 and Tom22 cannot be made. Also, the conclusion if an interaction is robust or not cannot be made as the pull-down lacks control fractions, it is also not clear how much of the eluate was loaded. Finally, Hfd1-HA was not expressed from its endogenous promoter, likely resulting in over-expression, which again strongly hampers conclusions about bona fide interaction partners.

We agree that our pulldown studies are done in an artificial context, such as Hfd1 overexpression needed for sufficient protein level for detection or use of Tapfusion proteins. However, the conclusion that Tom70 is a potential interactor of Hfd1 can be made based on the following observations: Hfd1-HA is preferentially pulled down from total protein extracts containing Tom70-Tap, but not from extracts containing no Tap-protein and significantly less from extracts containing Tom22-Tap, another TOM associated subunit. The pulldown assay has been repeated now several times and the efficiency of Hfd1 pulldown has been quantified and statistically analyzed with respect to the quantity of purified Tom protein, which is shown in modified Figure 8A.

Figure 4A and C: Depletion of proteasomal activity results in larger aggregates in Figure 4A. However, the addition of t-2-hex blocks proteasomal activity (Figure 4C). How can proteasome inhibition result in bigger aggregates if the proteasomal activity is lost upon t-2-hex addition?

The negative effect of t-2-hex on proteasomal activity is most likely an indirect effect caused by protein aggregation (Bence et al., Science 2001 292-1552) and occurs in wild type and rpn4 mutant cells with reduced proteasomal activity (Fig. 4C). t-2-hex causes cytosolic protein aggregation in wild type cells, which is aggravated (more and larger protein aggregates) in rpn4 mutants because of their lower levels of active proteasome (Fig. 4A). The observed protein aggregates will further diminish proteasomal activity, which is confirmed in Fig. 4C.

Figure 1B: The authors use a reporter to determine HFD1 expression that consists of the promoter region of HFD1 fused to luciferase. These fusion constructs have been shown to often not reflect the bona fide expression levels of genes (Yoneda et al., J Cell Sci 2004). qPCR analysis of transcript levels should be included to support the induction of HFD1.

We agree that the live cell luciferase reporters used here are not suitable for the determination of absolute mRNA levels. However, the aim of these reporter experiments is to quantify the inducibility of different genes (HFD1, GRE2) dependent on increasing stress doses. These dose response profiles cannot be obtained by qPCR analysis, while the

destabilized reporters are an excellent tool for this, which have been used to accurately describe numerous dynamic stress responses (for example: Dolz-Edo et al. 2013 MCB 33:2228-40, Rienzo et al. 2015 MCB 35:3669-83, PascualAhuir et al. 2019 BBA 862:457-471). Additionally, the induction of HFD1 mRNA levels by salt (NaCl) and oxidative (menadione) stress determined by qPCR has been published before (Manzanares-Estreded et al. 2017 Oxid Med Cell Longevity 2017:2708345).

The authors conclude from Figure 1 that entry into apoptotic cell death is modulated by efficient t-2-hex detoxification. However, this is based on growth curves and no analysis of apoptotic cell death is performed. The data show that the addition of hexadecenal results in a growth arrest, that is overcome likely upon degradation of t-2-hex (depending on the amount of Hfd1).

We agree that our experiments measure growth inhibition and not specifically apoptotic cell death. The text has been changed accordingly.

Figure 4A: Microscopy images show between 1-2 yeast cells. Either more cells need to be shown or quantifications of the aggregates are required. In addition, it is not clear if the control received the same DMSO concentration as the treated cells and also the time point for the control is not specified.

We have now quantified the number of aggregates across cell populations in new Figure 4A in DMSO, t-2-hex and t-2-hex-H2 treated wt and rpn4 mutants. These data show specific aggregate induction by t-2-hex and not by DMSO or the saturated t-2-hex-H2 control, which is aggravated in rpn4 mutants and avoided by CHX pretreatment.

Figure 5: Western blots in figure 5A, B, D, E and F lack a loading control. Without this, conclusions about increases in protein abundance cannot be made. Response: We have now included additional panels with the loading controls for the Western blots in new figure 5, except figure 5B, where the appearance or not of the pre-protein can be compared to the amount of mature protein in the same blot.

Figure 2B: Complex II assembly factors SDH5,6,9 are described here as ETC complexes. As the proteins are not part of the mature complex II, the heading should be modified into ETC complexes and ETC assembly.

Figure 2B has been revised and the classification of ETC proteins changed accordingly.

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